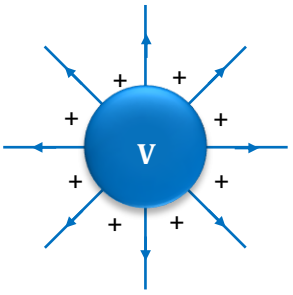
04 Capacitance

Introduction

A capacitor can store energy in the form of potential energy in an electric field. In this chapter we'll discuss the capacity of conductors to hold charge and energy.

Capacitance of an isolated conductor

When a conductor is charged its potential increases. For an isolated conductor (conductor should be of finite dimension, so that potential of infinity can be assumed to be zero) potential of the conductor is proportional to charge given to it.



q = charge on conductor

V = potential of conductor

$q \propto V \Rightarrow q = CV$

Here C is proportionality constant called capacitance of the conductor.

Illustration 1:

When $30\mu C$ charge is given to an isolated conductor of capacitance $5\mu F$. Find out the Potential of the conductor.

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Solution:

$$Q = 30\mu C$$

$$C = 5\mu F$$

$$V = \frac{Q}{C} = \frac{30}{5} = 6V$$

Illustration 2:

Two insulated conductors are charged by transferring electrons from one conductor to another. The potential difference of 100 V was produced on transferring 6.25×10^{15} electrons from one conductor to another. The capacity of the system will be.

Solution:

$$Q = CV \Rightarrow C = \frac{Q}{V} = \frac{ne}{V}$$

Given $V = 100$ volt ; $n = 6.25 \times 10^{15}$

$$\therefore \quad C = \frac{6.25\times10^{15}\times1.6\times10^{-19}}{100} = 10\mu F$$

Definition of capacitance:

Capacitance of conductor is defined as charge required to increase the potential of conductor by one unit.

Unit and dependency of capacitance of an isolated conductor:

(i) It is a scalar quantity.

(ii) Unit of capacitance is farad in SI units and its dimensional formula is $M^{-1}L^{-2}I^{2}T^{4}$.

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(iii) 1 Farad: 1 Farad is the capacitance of a conductor for which 1 coulomb charge increases potential by 1 volt.

$$1 \text{ Farad} = \frac{1 \text{ Coulomb}}{1 \text{ Volt}}$$

$1\,\mu F = 10^{-6}\,F, 1nF = 10^{-9}\,F$ or $1\,pF = 10^{-12}F$

(iv) Capacitance of an isolated conductor depends on following factors:

(a) Shape and size of the conductor: On increasing the size, capacitance increases.

(b) On surrounding medium: With increase in dielectric constant K, capacitance increases.

(c) Presence of other conductors: When a neutral conductor is placed near a charged conductor, capacitance of conductors increases.

(v) Capacitance of a conductor does not depend on

(a) Charge on the conductor

(b) Potential of the conductor

Capacitor

Definition

A capacitor or condenser consists of two conductors separated by an insulator or dielectric. Having opposite charges on which sufficient quantity of charge may be accommodated.

Principle

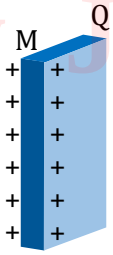
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It is based on the fact that if an identical earthed conductor is placed in the vicinity of a charged conductor then the capacitance of the charged conductor increases appreciable.

Consider a conducting plate M which is given a charge Q

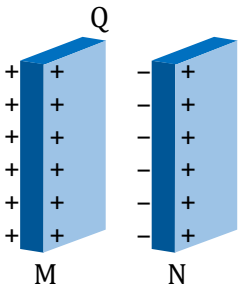
Such that its potential rises to V then,

$$C = \frac{Q}{V}$$



Let us place another identical conducting plate N parallel to it such that charge is induced on plate N (as shown in figure). If V_- is the potential at M due to induced negative charge on N and V_+ is the potential at M due to induced positive charge on N, then

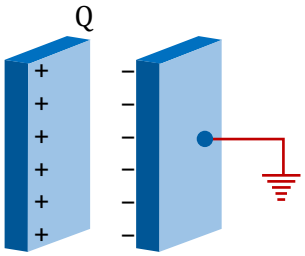
$$C' = \frac{Q}{V'} = \frac{Q}{V + V_+ - V_-}$$



Since $V' < V$ (as the induced negative charge lies closer to the plate M in comparison to induced positive charge). $\Rightarrow C' > C$ Further, if N is earthed from the outer side (see figure) then $V'' = V_+ - V_-$

($\because$ The entire positive charge flows to the earth)

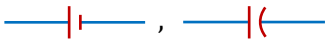
$$C'' = \frac{Q}{V''} = \frac{Q}{V - V_-} \Rightarrow C'' >> C'$$



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Circuit Representation:

The capacitor is represented as following:



Type

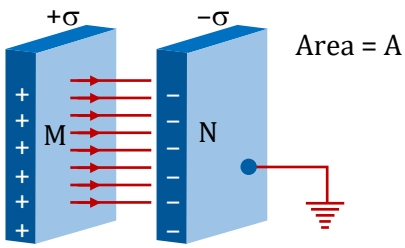
Based on shape and arrangement of capacitor plates there are various types of capacitors.

(a) Parallel plate capacitor (b) Spherical capacitor (c) Cylindrical capacitor

Parallel Plate Capacitor:

Capacitance

It consists of two metallic plates M and N each of area A at separation d. Plate M is positively charged and plate N is earthed. If ε_r is the dielectric constant of the material medium and E is the field at a point P that exists between the two plates, then



Step- I : Finding electric field,

$$E = E_+ + E_- = \frac{\sigma}{2\varepsilon} + \frac{\sigma}{2\varepsilon} = \frac{\sigma}{\varepsilon} = \frac{\sigma}{\varepsilon_0\varepsilon_r} \quad [\varepsilon = \varepsilon_0\varepsilon_r]$$

Step- II : Finding potential difference,

$$V = Ed = \frac{\sigma}{\varepsilon_0\varepsilon_r}\, d = \frac{qd}{A\varepsilon_0\varepsilon_r} \quad \left(\because E = \frac{V}{d} \text{ and } \sigma = \frac{q}{A}\right)$$

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Step- III : Finding capacitance,

$$C = \frac{q}{V} = \frac{\varepsilon_r\varepsilon_0 A}{d}$$

Dependency:

Capacitance of a capacitor depends on

(a) Area of plates. (b) Distance between the plates. (c) Dielectric medium between the plates.

Illustration 3:

The capacitance of a parallel plate capacitor is 10 μF when distance between its plates is 8 cm. If distance between the plates is reduced to 4 cm, its capacitance will be-

Solution:

$$C = \frac{\varepsilon_0 A}{d}; C' = \frac{\varepsilon_0 A}{d'}$$

$$d' = \frac{d}{2}$$

$$C' = 2C = 20\mu F$$

Illustration 4:

The capacitance of a parallel plate capacitor is $12\mu F$. If the distance between its plates is reduced to half and the area of plates is doubled, then the capacitance of the capacitor will become

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Solution:

$C = 12\ \mu F$

$d' = \frac{d}{2}$

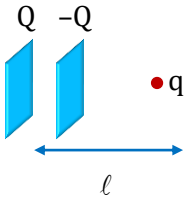
$A' = 2A$

$\frac{C'}{C} = \frac{A'}{A}\frac{d}{d'} = 2 \times 2 = 4$

$\Rightarrow \quad C' = 4C = 4 \times 12\mu F = 48\mu F$

Illustration 5:

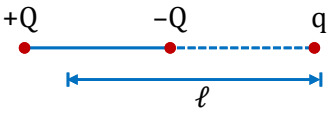
The plates of small size of a parallel plate capacitor are charged as shown. The magnitude of force on the charged particle of 'q' at a distance 'I' from the capacitor is : (Assume that the distance between the plates is $d << I$)



Solution:

The two plates acts as a dipole

$\therefore$ The magnitude of force on charge q ;



$$|\vec{F}| = |\vec{E}|q = \left(\frac{2kQd}{\ell^3}\right).q = \frac{Qqd}{2\pi\varepsilon_0\ell^3}$$

Types of Capacitors

Based on shape and arrangement of capacitor plates there can be various other types of capacitors :

(i) Spherical capacitor

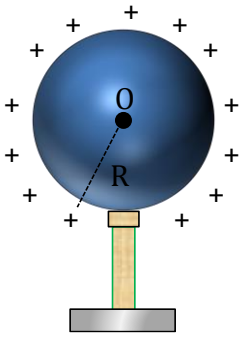
(ii) Cylindrical capacitor

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Spherical Capacitance

Capacitance of an Isolated Spherical Conductor

Let there is charge Q on sphere of radius R.



$\because$ Potential $V = \frac{KQ}{R}$

Hence by formula:

$$Q = CV \rightarrow Q = \frac{CKQ}{R} \rightarrow C = 4\pi\in_0 R$$

Capacitance of an isolated spherical conductor $\rightarrow C = 4\pi\in_0 R$

(a) If the medium around the conductor is vacuum or air.

$C_{vacuum} = 4\pi\in_0 R$ [R = Radius of spherical conductor. (may be solid or hollow.)]

If the medium around the conductor is a dielectric of constant K from surface of sphere to infinity then

$C_{medium} = 4\pi\in_0 KR$

(b) $\frac{C_{medium}}{C_{air/vaccum}} = K$ = dielectric constant

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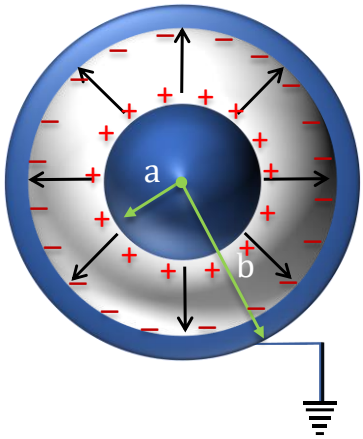
Illustration 6:

Find out the capacitance of the earth? (Radius of the earth = 6400 km)

Solution:

$$C = 4\pi\in_0 R = \frac{6400\times10^3}{9\times10^9} = 711\ \mu F$$

Capacitance of spherical capacitor



(a) Outer sphere is earthed

When a charge Q is given to inner sphere it is uniformly distributed on its surface, A charge $-Q$ is induced on inner surface of outer sphere. The charge $+Q$ induced on outer surface of outer sphere flows to earth as it is grounded.

$E = 0$ for $r < a$

and $E = 0$ for $r > b$

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This arrangement is known as spherical capacitor.

Potential of inner sphere, $V_1 = \frac{Q}{4\pi\varepsilon_0 a} + \frac{-Q}{4\pi\varepsilon_0 b} \Rightarrow \frac{Q}{4\pi\varepsilon_0}\left(\frac{b-a}{ab}\right)$

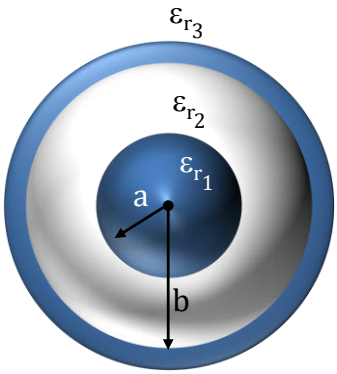
As outer surface is earthed so potential

$V_2 = 0$

$$V_1 - V_2 = \left[\frac{KQ}{a} - \frac{KQ}{b}\right] - \left[\frac{KQ}{b} - \frac{KQ}{b}\right] = \frac{KQ}{a} - \frac{KQ}{b}$$

$$C = \frac{Q}{V_1 - V_2} = \frac{Q}{\frac{KQ}{a} - \frac{KQ}{b}} = \frac{ab}{K(b-a)} = \frac{4\pi\in_0 ab}{b-a}$$

$$C = \frac{4\pi\in_0 ab}{b-a}$$



If $b >> a$ then $C = 4\pi\in_0 a$ (Like isolated spherical capacitor)

If dielectric mediums are filled as shown then: $C = \frac{4\pi\in_0\in_{r_2} ab}{b-a}$

Illustration 7:

The stratosphere acts as a conducting layer for the earth. If the stratosphere extends beyond 50 km from the surface of earth, then calculate the capacitance of the spherical capacitor formed between stratosphere and earth's surface. Take radius of earth as 6400 km.

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Solution:

The capacitance of a spherical capacitor is $C = 4\pi\varepsilon_0\left(\frac{ab}{b-a}\right)$

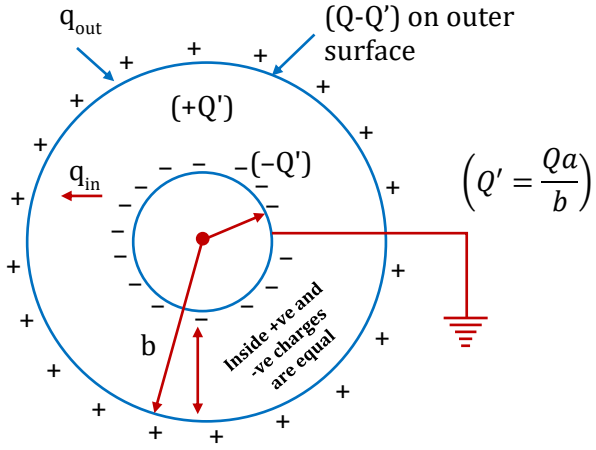
b = radius of the top of stratosphere layer = 6400 km + 50 km = 6450 km = 6.45 × 10^6 m

a = radius of earth = 6400 km = 6.4 × 10^6 m

$\therefore \quad C = \frac{1}{9\times10^9} \times \frac{6.45\times10^6\times6.4\times10^6}{6.45\times10^6 - 6.4\times10^6} = 0.092\ F$

(b) Inner sphere is earthed

Here the system is equivalent to a spherical capacitor of inner and outer radii a and b respectively and a spherical conductor of radius b in parallel. This is because charge Q given to outer sphere distributes in such a way that for the outer sphere charge on the inner side is $\frac{a}{b}Q$ and



Charge on the outer side is

$$Q - \frac{a}{b}Q = \frac{(b-a)}{b}Q$$

So total capacitance of the system

$$C = 4\pi\,\varepsilon_0\frac{ab}{b-a} + 4\,\pi\,\varepsilon_0\,b$$

$$\boxed{C = \frac{4\pi\varepsilon_0 b^2}{b-a}}$$

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Cyllindrical Capacitance

There are two co-axial conducting cylindrical surfaces where $\ell >> a$ and $\ell >> b$, where a and b are radius of cylinders.

When a charge Q is given to inner cylinder it is uniformly distributed on its surface. A charge $-Q$ is induced on inner surface of outer cylinder. The charge $+Q$ induced on outer surface of outer cylinder flows to earth as it is grounded.

Electrical field between cylinders, $E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{Q/\ell}{2\pi\varepsilon_0 r}$

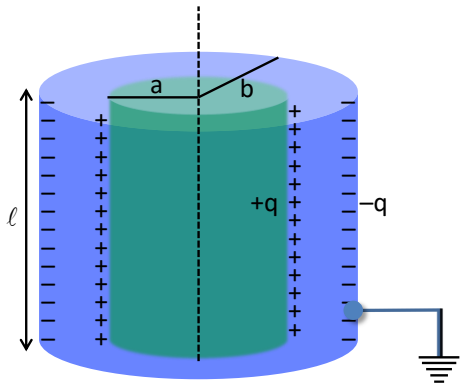
Potential difference between plates

$$V = \int_a^b \frac{Q}{2\pi\varepsilon_0 r\ell}dr = \frac{Q}{2\pi\varepsilon_0\ell}\ell n\left(\frac{b}{a}\right)$$

Capacitance per unit length

$$C = \frac{\lambda}{V} = \frac{\lambda}{2K\lambda\ell n\frac{b}{a}} = \frac{4\pi\epsilon_0}{2\ell n\frac{b}{a}} = \frac{2\pi\epsilon_0}{\ell n\frac{b}{a}}$$

Capacitance per unit length = $\frac{2\pi\epsilon_0}{\ell n\frac{b}{a}}\ F/m$



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Illustration 8:

A cylindrical capacitor has two co-axial cylinders of length 15 cm and radii 1.5 cm and 1.4 cm. The outer cylinder is earthed and the inner cylinder is given a charge of 3.5 μC. Determine the capacitance of the system and the potential of the inner cylinder.

Solution:

$\ell = 15$ cm $= 15 \times 10^{-2}\ m$; $a = 1.4$cm $= 1.4 \times 10^{-2}\ m$

$b = 1.5$ cm $= 1.5 \times 10^{-2} m$; $q = 3.5\ \mu C = 3.5 \times 10^{-6} C$

Capacitance $C = \dfrac{2\pi\varepsilon_0 \ell}{2.303\, log_{10}\left(\frac{b}{a}\right)} = \dfrac{2\pi \times 8.854 \times 10^{-12} \times 15 \times 10^{-2}}{2.303\, log_{10}\frac{1.5\times10^{-2}}{1.4\times10^{-2}}}$

$= 1.21 \times 10^{-10}\ F$

Since the outer cylinder is earthed, the potential of the inner cylinder will be equal to the potential difference between them. Potential of inner cylinder, is

$$V = \frac{q}{C} = \frac{3.5\times10^{-6}}{1.2\times10^{-10}} = 2.89 \times 10^4 V$$

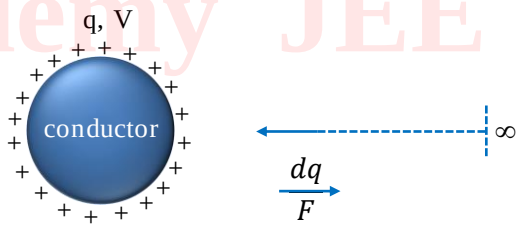
Work done by external agent to charge a conductor

Work has to be done to charge a conductor because charge present on it will repel the next incoming charged element.

Let at any instant charge of conductor is q and potential is V.

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Work done by external agent to bring small charged element dq from infinity to surface of conductor.



$dw = dq\ [V_f - V_i]$, $V_i = V_\infty = 0$, $V_f = V$

$dw = dq\ [V - 0]$

$dw = V\, dq$

$\int dw = \int V.dq$

$\because$ $V = q/C$

Work done by external agent to charge the conductor from q_1 to q_2 is

$$W = \int_{q_1}^{q_2} \frac{q}{C}.dq \quad \Rightarrow \quad W = \frac{1}{C}\left(\frac{q^2}{2}\right)_{q_1}^{q_2} \Rightarrow W = \frac{q_2^2 - q_1^2}{2C}$$

$q_1 = CV_1$, $q_2 = CV_2$

Work done by external agent to increase the potential of conductor from V_1 to V_2

$$W = \frac{C}{2}(V_2^2 - V_1^2)$$

For 0 to V

$$w = \frac{C}{2}(V^2 - 0^2) = \frac{1}{2}CV^2$$

So, $w = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C}$

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Potential energy of a charged conductor

Since electrostatic field is conservative field and in conservative field work done by external agent is stored in form of potential energy.

* Potential energy of conductor will be stored in form of electric field.
* Potential energy of a conductor which is charged by V potential is given by

$P.E. = W_{ext}$, $\quad U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$

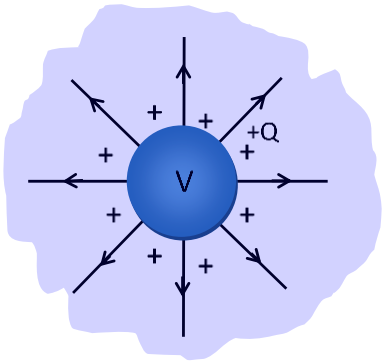
Potential Energy of Conducting Sphere:

$$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$$

$\because \quad C = 4\pi\varepsilon_0 R$

$$U = \frac{1}{2}\frac{Q^2}{4\pi\varepsilon_0 R}$$

$$U = \frac{KQ^2}{2R}$$



Self-potential energy of spherical conductor.

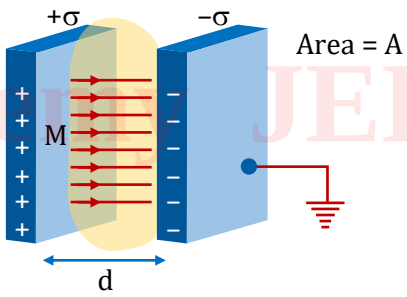
This potential energy is stored in the form of electric field.

Energy of parallel plate capacitor:

Energy stored in capacitor = $\frac{Q^2}{2C}$ putting the value of capacitance for parallel plate capacitor

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{(A\sigma)^2}{2} \times \frac{d}{\varepsilon_0 A}$$

$$U = \frac{1}{2}(Ad)\frac{\sigma^2}{\epsilon_0}$$



Energy density

$$\frac{U}{V} = \frac{\sigma^2}{2\epsilon_0} = \frac{1}{2}\epsilon_0 E^2$$ (here V is volume i.e. A.d and $E = \frac{\sigma}{\epsilon_0}$)

Once it is established that a region containing electric field E has energy $\frac{1}{2}\epsilon_0 E^2$ per unit volume, the result can be used for any electric field whether it is due to a capacitor or otherwise.

Illustration 9:

The work done against electric forces in increasing the potential difference of a condenser from 20 V to 40 V is W. The work done in increasing its potential difference from 40 V to 50 V will be

Solution:

$$W = U_f - U_i = \frac{1}{2}CV_f^2 - \frac{1}{2}CV_i^2 = \frac{1}{2}C(40^2 - 20^2)\ W = 600\ C$$

$$W_1 = \frac{1}{2}C(50^2 - 40^2) = \frac{900}{2}C$$

$$W_1 = \frac{900}{2} \cdot \frac{W}{600} = \frac{3}{4}W$$

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Illustration 10:

The plate separation in a parallel plate condenser is d and plate area is A. If it is charged to V volt and battery is disconnected then the work done in increasing the plate separation to $2d$ will be:

Solution:

As battery is disconnected, charge remains constant in the work process.

Work done = final potential energy – initial potential energy

$$= \frac{Q^2}{2C'} - \frac{Q^2}{2C^0} = \frac{Q^2}{2}\left\{\frac{1}{C'} - \frac{1}{C^0}\right\}$$

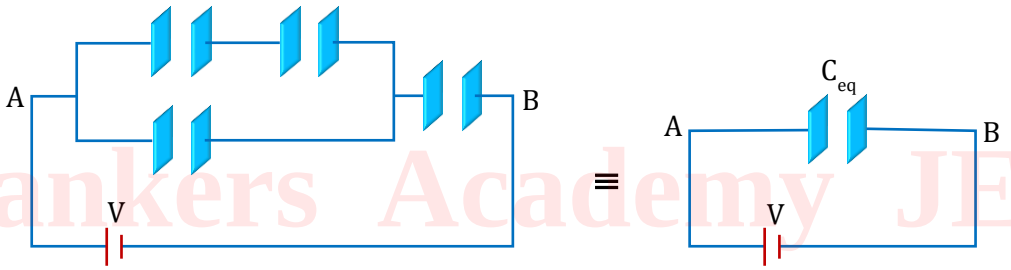
Where, $Q = CV = \frac{A\varepsilon_0 V}{d}$; $C = \frac{A\varepsilon_0}{d}$; $C' = \frac{A\varepsilon_0}{2d}$

Now, work done = $\frac{\varepsilon_0 V^2 A}{2d}$

Meaning of Equivalent Capacitor:

(a) Replacing the given capacitor network between two terminals by a single capacitor so that this single capacitor produces the same electrical effect when connected between the same two terminals instead of the original capacitors network.

(b) Capacitance of this single capacitor which can be used to replace the original capacitor network is termed as equivalent capacitance of the given network.



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$$C_{eq} = \frac{Q}{V}$$

(c) Here same electrical effect refers to the charge flown through the battery or charge transferred between the terminals plates of capacitors connected to battery. It could also be understood in form of energy stored being same in two cases i.e. when original capacitor network is connected and the one when original network is replaced by equivalent capacitance.

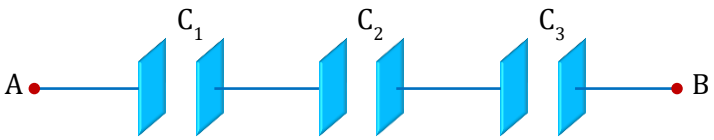
There are two standard combinations:

(1) Series Combination (2) Parallel Combination

Series Combination

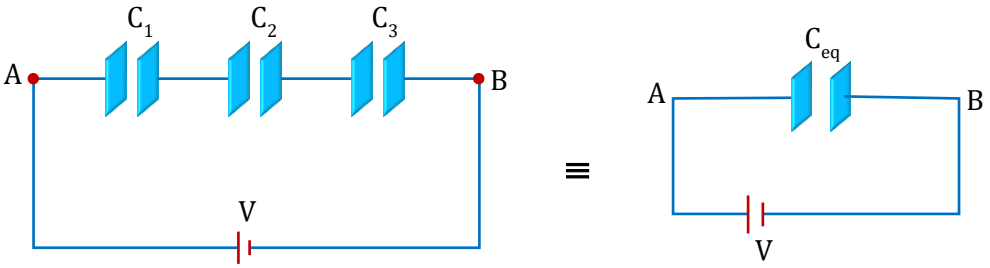
When initially uncharged capacitors are connected as shown so that charges do not have any alternative path(s) to flow then the combination is called series combination.

Derivation:

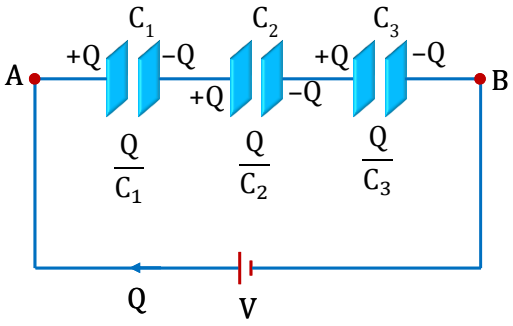


To find equivalent capacitance of this combination lets connect a battery across its terminals.

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$$C_{eq} = \frac{Q}{V}$$



Let's assume that initially, the capacitors were uncharged and after connecting to battery, Q charge flows through the battery as shown in above figure.

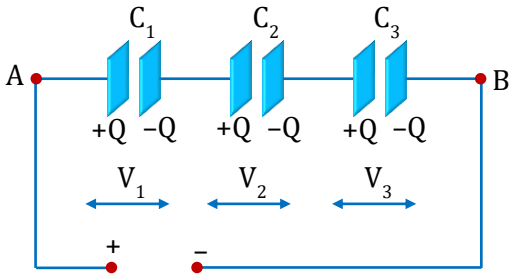
Applying Kirchhoff's voltage law

$$\frac{-Q}{C_1} + \frac{-Q}{C_2} + \frac{-Q}{C_3} + V = 0. \; ; \; \frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\Rightarrow \quad \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \text{ or in general } \frac{1}{C_{eq}} = \sum_{n=1}^{n} \frac{1}{C_n}$$

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Salient features of series combination



(a) All capacitors will have same charge but can have different potential difference across them.

(b) We can say that potential difference across capacitor is inversely proportional to its capacitance in series combination $\propto \frac{1}{C}$

$$V_1 = \frac{Q}{C_1}, V_2 = \frac{Q}{C_2}, V_3 = \frac{Q}{C_3} \ldots$$

(c) $V_1 : V_2 : V_3 = \frac{1}{C_1} : \frac{1}{C_2} : \frac{1}{C_3}$

In series combination the smallest capacitor gets maximum potential.

(d) $$V_1 = \frac{\frac{1}{C_1}}{\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}+\ldots\ldots} V \quad V_2 = \frac{\frac{1}{C_2}}{\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}+\ldots\ldots} V$$

$$V_3 = \frac{\frac{1}{C_3}}{\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}+\ldots.} V$$

Where $V = V_1 + V_2 + V_3$

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(e) In series : $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$

In series combination equivalent capacitance is always less the smallest capacitor of combination.

(f) Energy stored in the combination $U_{\text{combination}} = \frac{Q^2}{2C_1} + \frac{Q^2}{2C_2} + \frac{Q^2}{2C_3}$

$U_{\text{combination}} = \frac{Q^2}{2C_{eq}}$

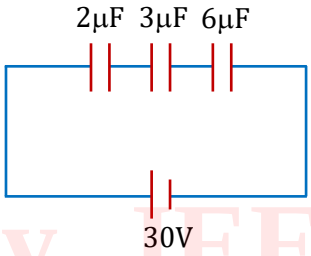
Energy supplied by the battery in charging the combination $U_{\text{battery}} = Q \times V = Q \cdot \frac{Q}{C_{eq}} = \frac{Q^2}{C_{eq}}$

$\frac{U_{\text{combination}}}{U_{\text{battery}}} = \frac{1}{2}$

Half of the energy supplied by the battery is stored in form of electrostatic energy and half of the energy is converted into heat through resistance. (if capacitors are initially uncharged)

Illustration 11:

Three initially uncharged capacitors are connected in series as shown in circuit with a battery of emf 30 V. Find out following:



(i) charge flow through the battery

(ii) potential energy in 3μF capacitor

(iii) U_{total} in capacitors

(iv) heat produced in the circuit

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Solution:

$\frac{1}{C_{eq}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = 1$

$C_{eq} = 1\mu F$

(i) $Q = C_{eq} V = 30\ \mu C$

(ii) Charge on 3μF capacitor = 30 μC

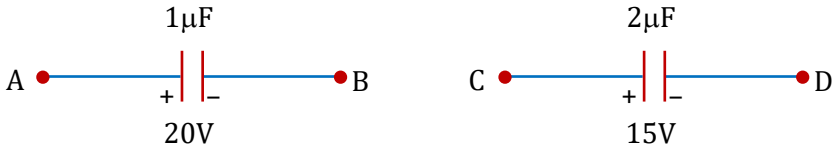
Energy = $\frac{Q^2}{2C} = \frac{30\times30}{2\times3} = 150\mu J$

(iii) $U_{\text{total}} = \frac{30\times30}{2}\mu J = 450\ \mu J$

(iv) Heat produced = (30 μC) (30) – 450 μJ = 450 μJ.

Illustration 12:

Two capacitors of capacitance 1 μF and 2μF are charged to potential difference 20 V and 15 V as shown in figure. If now terminal B and C are connected together terminal A with positive of battery and D with negative terminal of battery 30 V then find out final charges on both the capacitor.



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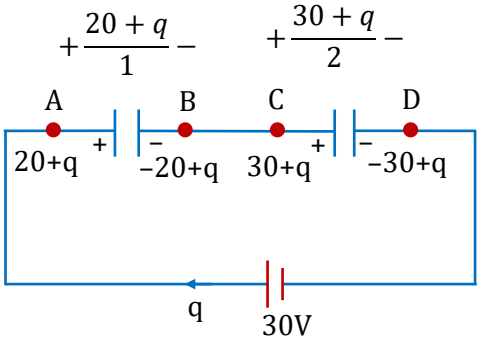
Solution:

Now applying Kirchoff voltage law $\frac{-(20+q)}{1} - \frac{30+q}{2} + 30 = 0$

$-40 - 2q - 30 - q = -60$

$3q = -10$

Charge flow $= -10/3\ \mu C$.



Charge on capacitor of capacitance $1\mu F = 20 + q = \frac{50}{3}\mu C$

Charge on capacitor of capacitance $2\mu F = 30 + q = \frac{80}{3}\mu C$

Parallel Combination:

When one plate of each capacitors (more than one) is connected together and the other plate of each capacitor is connected together, such combination is called parallel combination.

Derivation of Equivalent Capacitance

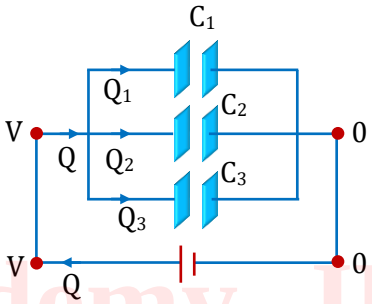
$Q = Q_1 + Q_2 + Q_3$

$Q = C_1V + C_2V + C_3V = V(C_1 + C_2 + C_3)$

$\frac{Q}{V} = C_1 + C_2 + C_3$

$C_{eq} = C_1 + C_2 + C_3$

In general $C_{eq} = \sum_{n=1}^{n} C_n$



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Salient Features of Parallel Combination

(a) All capacitors have same potential difference but can have different charges.

(b) We can say that:

$Q_1 = C_1V$

Q_1 = Charge on capacitor C_1

C_1 = Capacitance of capacitor C_1

V = Potential across capacitor C_1

- $Q_1 : Q_2 : Q_3 = C_1 : C_2 : C_3$

  The charge on the capacitor is proportional to its capacitance

  $Q \propto C$

  $Q_1 = \frac{C_1}{C_1+C_2+C_3}Q \Rightarrow Q2 = \frac{C_2}{C_1+C_2+C_3}Q$

  $Q_3 = \frac{C_3}{C_1+C_2+C_3}Q$

  Where $Q = Q_1 + Q_2 + Q_3$ .....

  Maximum charge will flow through the capacitor of largest value.

- $C_{eq} = C_1 + C_2 + C_3$

  Equivalent capacitance is always greater than the largest capacitor of combination.

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- Energy stored in the combination :

$$U_{\text{combination}} = \frac{1}{2}C_1V^2 + \frac{1}{2}C_2V^2 + \cdots = \frac{1}{2}(C_1 + C_2 + C_3 \ldots\ldots)V^2 = \frac{1}{2}C_{eq}V^2$$

$$U_{\text{battery}} = QV = C_{eq}V^2$$

$$\frac{U_{\text{combination}}}{U_{\text{battery}}} = \frac{1}{2}$$

Half of the energy supplied by the battery is stored in form of electrostatic energy and half of the energy is converted into heat through resistance. (If all capacitors are initially uncharged)

Illustration 13:

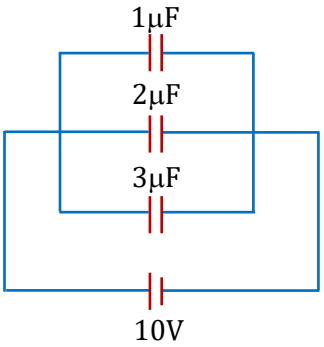
Three initially uncharged capacitors are connected to a battery of 10 V is parallel combination find out following :

(i) charge flown from the battery

(ii) total energy stored in the capacitors

(iii) heat produced in the circuit

(iv) potential energy in the 3μF capacitor.



Solution:

(i) $Q = C_{eq}V = (3 + 2 + 1)10 = 60\ \mu C$ (ii) $U_{\text{total}} = \frac{1}{2} \times 6 \times 10 \times 10 = 300\ \mu J$

(iii) heat produced = $60 \times 10 - 300 = 300\ \mu J$ (iv) $U_{3\mu F} = \frac{1}{2} \times 3 \times 10 \times 10 = 150\ \mu J$

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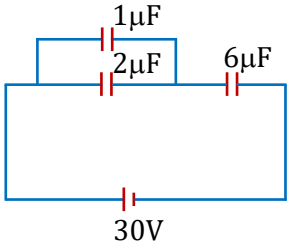
Mixed Combination:

The combination which contains mixing of series parallel combinations or other complex combinations fall in mixed category. There are two types of mixed combinations

(i) Simple (ii) Complex

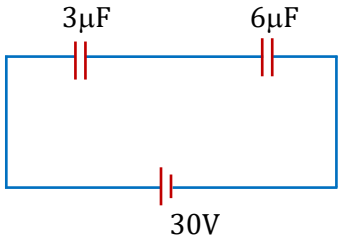
Illustration 14:

In the given circuit find out charge on 6μF and 1μF capacitor.



Solution:

It can be simplified as $C_{eq} = \frac{18}{9} = 2\mu F$



Charge flow through the cell = $30 \times 2\mu C$

$Q = 60\mu C$

Now charge on 3μF = Charge on 6μF= 60μC

Potential difference across 3μF = 60/ 3= 20V

∴ Charge on 1μF = 20 μC

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Illustration 15:

An infinite number of capacitors of capacitance C, $4C$, $16C$... ∞ are connected in series then what will be their resultant capacitance?

Solution:

Let the equivalent capacitance of the combination $= C_{eq}$

$$\frac{1}{C_{eq}} = \frac{1}{C} + \frac{1}{4C} + \frac{1}{16C} + \ldots \infty = \left[1 + \frac{1}{4} + \frac{1}{16} + \ldots \infty\right] \frac{1}{C} \text{ (this is G.P. series)}$$

$\Rightarrow$ $S_\infty = \frac{a}{1-r}$ first term $a = 1$, common ratio $r = \frac{1}{4}$

$\Rightarrow$ $\frac{1}{C_{eq}} = \frac{1}{1-\frac{1}{4}} \times \frac{1}{C} \Rightarrow C_{eq} = \frac{3}{4}C$

Combination of Parallel Plates:

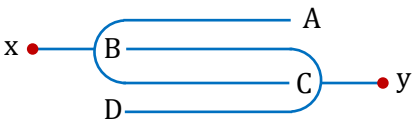
Combinations of parallel plates can be grouped and equivalent capacitance can be found easily by redrawing the circuit by connecting plates to common terminals. This has been illustrated by following example.

Four parallel plates each of area A and plate separation d. These four plates are forming three parallel plate capacitors, one between plates A and B, one between plates B and C and another between plates C and D. Capacitance of each of these capacitors is given as $C = \frac{\epsilon_0 A}{d}$.

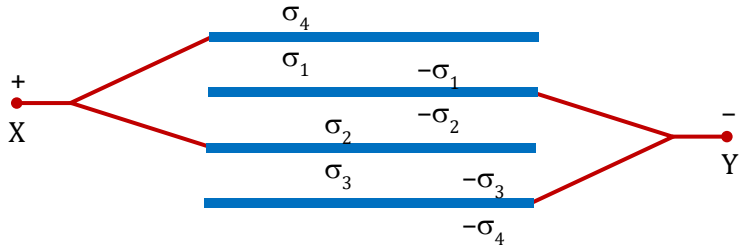
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If plates A and C are connected as terminal X and plates B and D are connected as terminal Y. Then calculation of equivalent capacitance across terminal X and Y can be done by following method.

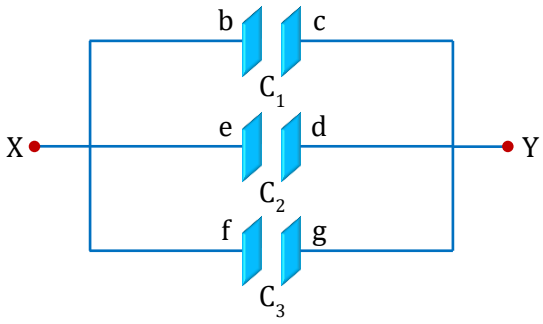


Let us assume that we have connected a battery between terminal X and Y. This battery supplies equal and opposite charges to terminal X and Y. For the detailed analysis let us say σ_1 and σ_2 are charge density on faces b and e. Similarly, σ_3 and σ_4 are charge density on faces f and a.



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Now equal opposite charges will be there on opposite faces as shown in figure (as per Gauss Law). Also electric field at any point inside conductor must be zero therefore σ_4 must be zero.

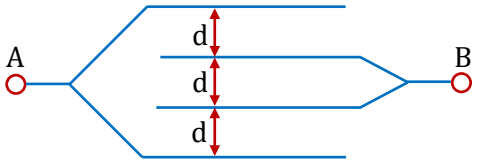


$V_{AB} = V_{BC} = V_{CD}$

So, all the capacitors are in parallel combination.

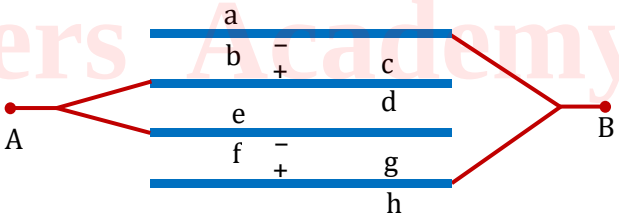
$$C_{eq} = C_1 + C_2 + C_3 = \frac{3A\epsilon_0}{d}$$

Let us try to find out equivalent capacitance between A and B. (take each plate Area = A) in following case by using the same above concept.

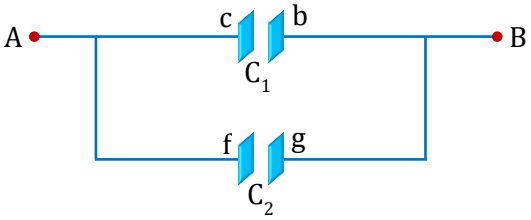


Assuming a battery between terminals A and B the positive and negative charges on the plates will be as shown in the figure fellow.

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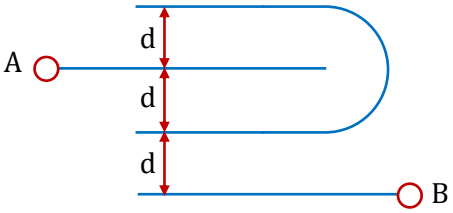
Here charges on plates d and e is zero because their potential must be same, so no electric field should be there in between these faces, so charge on faces d and e must be zero.



These are only two capacitors, $C_{eq} = C_1 + C_2 = \frac{2A\epsilon_0}{d}$

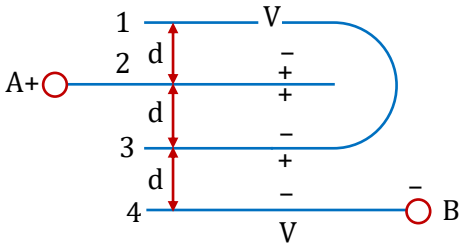
Illustration 16:

Find out equivalent capacitance between A and B. (take each plate Area = A)



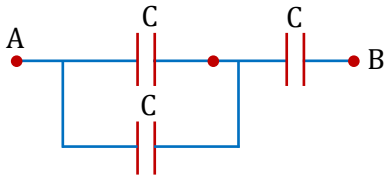
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Solution:



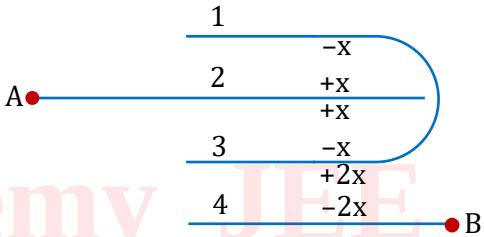
The modified circuit is

$$C_{eq} = \frac{2C}{3} = \frac{2A\epsilon_0}{3d}$$



Other method:

Let charge density as shown



$$C_{eq} = \frac{Q}{V} = \frac{2xA}{V}$$

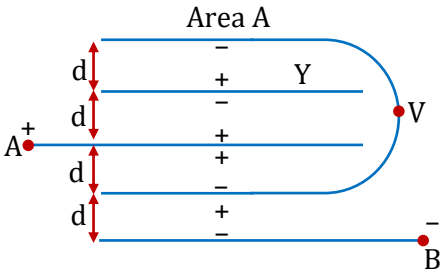
$$V = V_2 - V_4 = (V_2 - V_3) + (V_3 - V_4) = \frac{xd}{\epsilon_0} + \frac{2xd}{\epsilon_0} = \frac{3xd}{\epsilon_0}$$

$$\therefore \quad C_{eq} = \frac{2Ax\,\epsilon_0}{3xd} = \frac{2A\,\epsilon_0}{3d} = \frac{2C}{3}.$$

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Illustration 17:

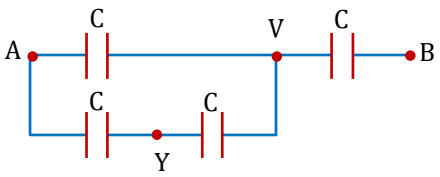
Find out equivalent capacitance between *A* and *B*.



Solution:

Let $C = \frac{A\epsilon_0}{d}$

Equivalent circuit :



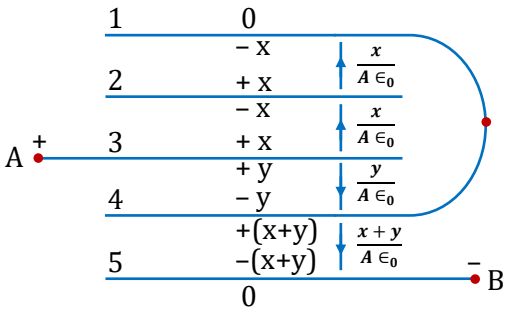
$$\frac{1}{C_{eq}} = \frac{1}{C} + \frac{2}{3C} = \frac{5}{3C}$$

$$C_{eq} = \frac{3C}{5} = \frac{3A\,\epsilon_0}{5d}$$

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Alternative Method:

Let charge distributions on plates as shown:



$$C = \frac{Q}{V} = \frac{x+y}{V_{AB}}$$

Potential of 1 and 4 is same

$$\frac{y}{A\epsilon_0} = \frac{2x}{A\epsilon_0} \qquad y = 2x$$

$$V = \left(\frac{2y+x}{A\varepsilon_0}\right)d$$

$$\Rightarrow \qquad C = \frac{(x+2x)A\epsilon_0}{(5x)d} = \frac{3A\epsilon_0}{5d}$$

The two plates of capacitor attract each other because they are oppositely charged.

Electric field due to positive plate,

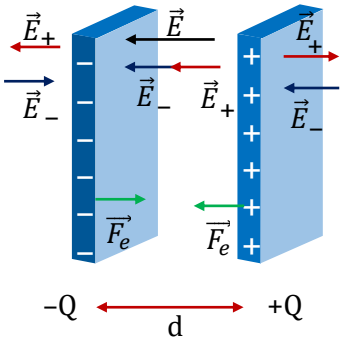
$$E = \frac{\sigma}{2\varepsilon_0} = \frac{Q}{2\varepsilon_0 A}$$

Force on negative charge $-Q$ is

$$F = -QE = -\frac{Q^2}{2\varepsilon_0 A}$$

Magnitude of force

$$F = \frac{Q^2}{2\varepsilon_0 A} = \frac{1}{2}\,\varepsilon_0 A E^2$$



Force per unit area or energy density or electrostatic pressure

$$= \frac{F}{A} = u = p = \frac{1}{2}\,\epsilon_0 E^2$$

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Note:

Electric field between the plates of a capacitor is shown in figure. Non-uniformity of electric field at the boundaries of the plates is negligible if the distance between the plates is very small as compared to the length of the plates.

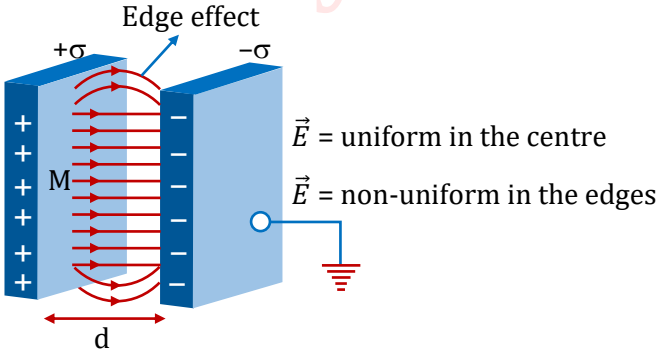
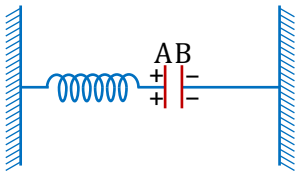


Illustration 18:

Plate A of a parallel air-filled capacitor is connected to a non-conducting spring having force constant k and plate B is fixed. If a charge $+q$ is placed on plate A and charge $-q$ on plate B then find out extension in the spring in equilibrium. Assume area of plate is 'A'.



Solution:

$$F_{\text{attraction}} = \frac{q^2}{2\epsilon_0 A};\ F_{\text{Repulsion}} = kx$$

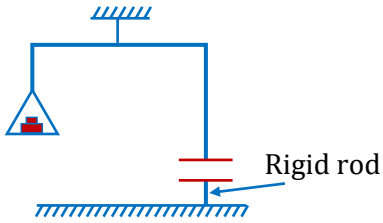
$$|F_{\text{attraction}}| = |F_{\text{Repulsion}}|$$

$$x = \frac{q^2}{2k\epsilon_0 A}$$

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Illustration 19:

The lower plate of a parallel plate capacitor is supported on a rigid rod. The upper plate is suspended from one end of a balance. The two plates are joined together by a thin wire and subsequently disconnected. The balance is then counterpoised. Now a voltage $V = 5000$ volt is applied between the plates. The distance between the plates is $d = 5\ mm$ and the area of each plate is $A = 100\ cm^2$. Then find out the additional mass placed to maintain balance.



[All the elements other than plates are massless and non-conducting]

Solution:

The electric force between the plates will be balanced by the additional weight

Hence $mg = \frac{Q^2}{2A\epsilon_0} = \frac{C^2V^2}{2A\epsilon_0}$

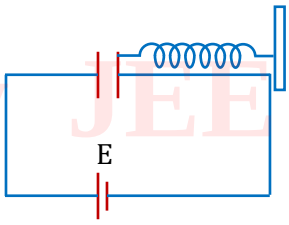
$$mg = \frac{\epsilon_0 AV^2}{2d^2}$$

$$m = \frac{\epsilon_0\, AV^2}{2d^2 g} = \frac{\epsilon_0 \times 100 \times 10^{-4}(5000)^2}{2(5\times 10^{-3})^2 \times 10}$$

$m = 4.425\ g$ **Ans.**

Illustration 20:

One plate of a capacitor is fixed and the other is connected to a spring as shown in Fig. Area of both the plates is A. In steady state (equilibrium), separation between the plates is $0.8\ d$ (spring was unstretched and the distance between the plates was d when the capacitor was uncharged). The force constant of the spring is approximately.



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Solution:

$$\frac{Q^2}{2A\varepsilon_0} = K(0.2d)$$

$$\frac{\left(\frac{\varepsilon_0 A}{0.8d}E\right)^2}{2A\varepsilon_0} = 0.2Kd$$

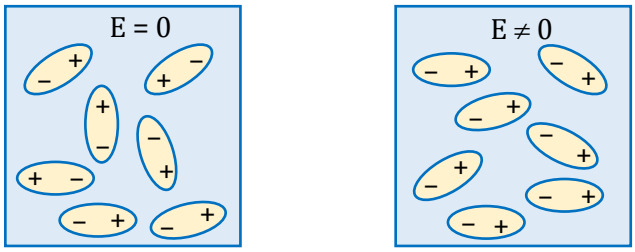
$$K = \frac{3.9\varepsilon_0 AE^2}{d^3}$$

Effect of Dielectric

- The insulators in which at microscopic level displacement of charges takes place in presence of electric field are known as **dielectrics**.
- Dielectrics are non-conductors upto certain value of field depending on its nature. If the field exceeds this limiting value called **dielectric strength** they lose their insulating property and begin to conduct.
- **Dielectric strength** is defined as the maximum value of electric field that a dielectric can tolerate without breakdown. Unit is volt/metre. Dimensions $M^1\ L^1\ T^{-3}\ A^{-1}$.

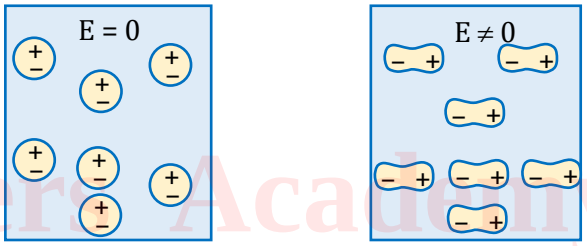
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Polar dielectrics



- In absence of external field, the centres of positive and negative charge do not coincide-due to asymmetric shape of molecules.
- Each molecule has permanent dipole moment.
- The dipole are randomly oriented so average dipole moment per unit volume of polar dielectric in absence of external field is nearly zero.
  In presence of external field dipoles tends to align in direction of field.
  Example: Water, Alcohol CO_2, $HC\ell$, NH_3

Non polar dielectrics



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- In absence of external field the centre of positive and negative charge coincides in these atoms or molecules because they are symmetric.
- The dipole moment is zero in normal state.
- In presence of external field they acquire induced dipole moment.
  Example: Nitrogen, Oxygen, Benzene, Methane

Polarisation:

The alignment of dipole moments of permanent or induced dipoles in the direction applied electric field is called polarisation.

- **Polarisation Vector $\vec{P}$**
  This is a vector quantity which describes the extent to which molecules of dielectric become polarized by an electric field or oriented in direction of field.
  $\vec{P}$ = the dipole moment per unit volume of dielectric = $n\vec{p}$
  where n is number of atoms per unit volume of dielectric and P is dipole moment of an atom or molecule.
  $\vec{P} = n\vec{p} = \frac{q_i d}{Ad} = \left(\frac{q_i}{A}\right) = \sigma_i$ = induced surface charge density.

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Capacitors with Dielectric:

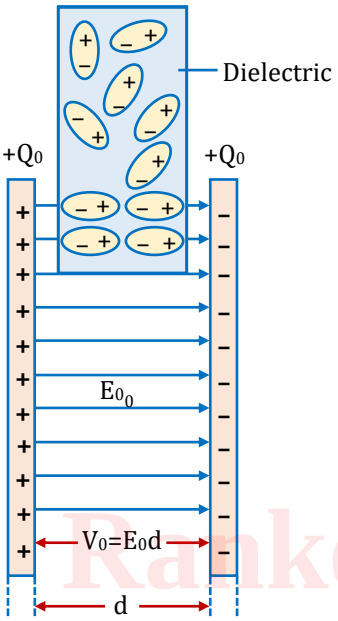
(i) In absence of dielectric $E = \frac{\sigma}{\epsilon_0}$

(ii) When a dielectric fills the space between the plates then molecules having dipole moment align themselves in the direction of electric field.

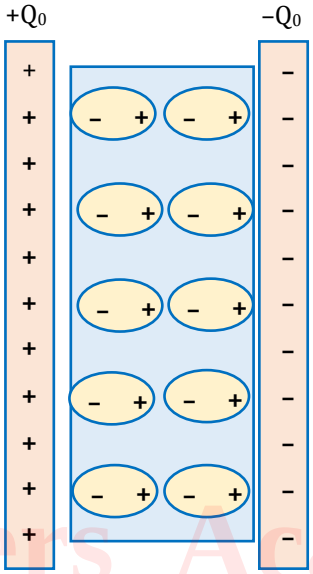
σ_b = induced (bound) charge density (called bound charge because it is not due to free electrons).

The induced charge also produce electric field.

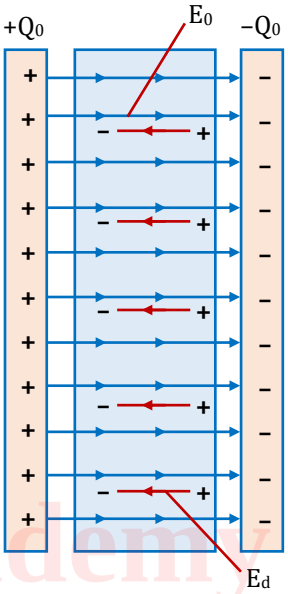
Let E_0, V_0, C_0 be electric field, potential difference and capacitance in absence of dielectric and E, V, C are electric field, potential difference and capacitance respectively in presence of dielectric.



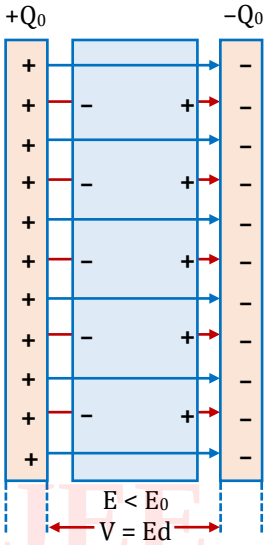
Charged capacitor



Charged capacitor
With dielectric inserted



Electric field
Inside dielectric



Effect on electric field
and voltage

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Electric field in absence of dielectric, $E_0 = \frac{V_0}{d} = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$

Electric field in presence of dielectric, $E = E_0 - E_i = \frac{\sigma - \sigma_b}{\varepsilon_0} = \frac{Q - Q_b}{\varepsilon_0} = \frac{V}{d}$

Capacitance in absence of dielectric, $C_0 = \frac{Q}{V_0}$

Capacitance in presence of dielectric, $C = \frac{Q - Q_b}{V}$

The dielectric constant or relative permittivity K or $\varepsilon_r = \frac{E_0}{E} = \frac{V_0}{V} = \frac{C}{C_0} = \frac{Q}{Q - Q_b} = \frac{\sigma}{\sigma - \sigma_b} = \frac{\varepsilon}{\varepsilon_0}$

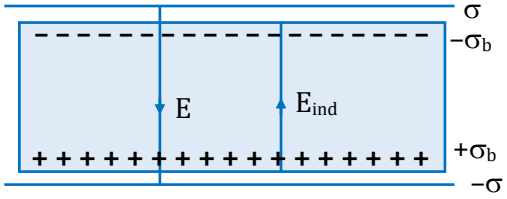
From $K = \frac{Q}{Q - Q_b}, Q_b = Q\left(1 - \frac{1}{K}\right)$ and $K = \frac{\sigma}{\sigma - \sigma_b} \Rightarrow \sigma_b = \sigma\left(1 - \frac{1}{K}\right)$

(iii) Capacitance in the presence of dielectric

$$C = \frac{\sigma A}{V} = \frac{\sigma A}{\frac{\sigma}{K\epsilon_0}.d} = \frac{AK\epsilon_0}{d}$$

Here capacitance is increased by a factor K.

$$C = \frac{AK\epsilon_0}{d}$$



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Illustration 21:

A parallel-plate capacitor is formed by two plates, each of area 100 cm^2, separated by a distance of 1mm. A dielectric of dielectric constant 5.0 and dielectric strength 1.9×10^7 V/m is filled between the plates. Find the maximum charge that can be stored on the capacitor without causing any dielectric breakdown.

Solution:

If the charge on the capacitor = Q

the surface charge density $\sigma = \frac{Q}{A}$ and the electric field $= \frac{Q}{KA\varepsilon_0}$.

This electric field should not exceed the dielectric strength 1.9×10^7 V/m.

$\therefore$ If the maximum charge which can be given is Q

then $\frac{Q}{KA\varepsilon_0} = 1.9 \times 10^7$ V/m, $A = 100\ cm^2 = 10^{-2}\ m^2$

$\Rightarrow$ $Q = (5.0) \times (10^{-2}) \times (8.85 \times 10^{-12}) \times (1.9 \times 10^7) = 8.4 \times 10^{-6}$ C.

Illustration 22:

Hard rubber has a dielectric constant of 2.8 and a dielectric strength (maximum electric field) of 18×10^6 volt/meter. If it is used as the dielectric material filling the full space in a parallel plate capacitor. What minimum area may the plates of the capacitor have in order that the capacitance be 7.0×10^{-2} μF and that the capacitor be able to withstand a potential difference of 4000 volts. ($\in_0 = 8.85 \times 10^{-12}$ S.I unit)

Solution:

Let distance between the plates = d

$18 \times 10^6 \times d = 4000$

$d = \frac{4000}{18 \times 10^6}$

Now, $C = \frac{\in_r A \in_o}{d}$

$7.0 \times 10^{-2} \times 10^{-6} = \frac{8.85 \times 10^{-12} \times A \times 2.8}{\frac{4000}{18 \times 10^6}}$

Solving, we get $A = 0.62\ m^2$.

Illustration 23:

A capacitor has two circular plates whose radius are 8cm and distance between them is 1mm. When mica (dielectric constant = 6) is placed between the plates, calculate the capacitance of this capacitor and the energy stored when it is given potential of 150 volt.

Solution:

Area of plate $\pi r^2 = \pi \times (8 \times 10^{-2})^2 = 0.0201\ m^2$ and $d = 1mm = 1 \times 10^{-3}\ m$

Capacity of capacitor,

$C = \frac{\varepsilon_0 \varepsilon_r A}{d} = \frac{8.85 \times 10^{-12} \times 6 \times 0.0201}{1 \times 10^{-3}} = 1.068 \times 10^{-9}\ F$

Potential difference,

$V = 150$ volt

Energy stored,

$U = \frac{1}{2} CV^2 = \frac{1}{2} \times (1.068 \times 10^{-9}) \times (150)^2 = 1.2 \times 10^{-5}\ J$

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Illustration 24:

A capacitor with air as the dielectric is charged to a potential of 100 volts. If the space between the plates is now filled with a dielectric of dielectric constant 10, the potential difference (in volt) between the plates will be:

Solution:

$V = 100\ V,\quad K = 10$

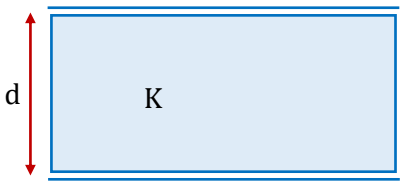
$Q = CV$ (constant), $C' = KC$

$V' = \frac{V}{K} = 10V$

Capacitor with Dielectric in Different Configuration

Calculation of Capacitance

When **single** dielectric is completely filled inside capacitor



$$C = \frac{\sigma A}{V} = \frac{\sigma A}{\frac{\sigma}{K \in_0}.d} = \frac{AK \in_0}{d}$$

Here capacitance is increased by a factor K.

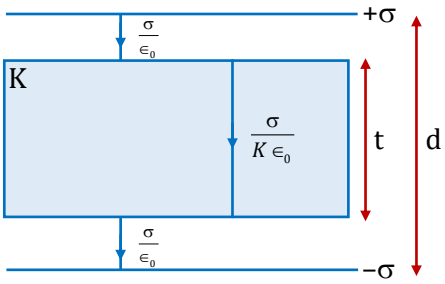
$$C = \frac{AK \in_0}{d}$$

Now lets try to understand how to calculate capacitance in other configurations.

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Single or multiple dielectric is partially filled

When the dielectric is filled partially between plates, the thickness of dielectric slab is t ($t < d$). If the medium does not fill between the plates completely then electric field will be as shown in figure.



The total electric field produced by bound induced charge on the dielectric outside the slab is zero because they cancel each other and hence electric field will be only due to plate charges in these vacant spaces. Where as electric field in the dielectric space will be vector sum of electric field due to plate charges and induced charges as shown in figure.

Then $V = E_0\,(d - t) + Et$

If V is total potential,

$\Rightarrow \quad V = E_0\left[d - t + \left(\frac{E}{E_0}\right)t\right]\frac{E_0}{E} = \varepsilon_r$ = Dielectric constant

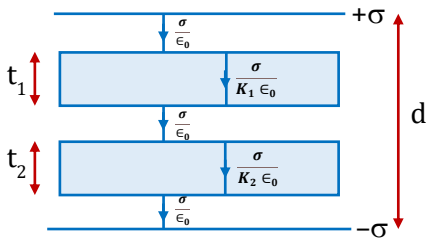
$\Rightarrow \quad V = \frac{\sigma}{\varepsilon_0}\left[d - t + \frac{t}{\varepsilon_r}\right] = \frac{q}{A\varepsilon_0}\left[d - t + \frac{t}{\varepsilon_r}\right]$

$\Rightarrow \quad C = \frac{q}{V} = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{\varepsilon_r}\right)}$

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Example:

If v is the potential difference between the plate,

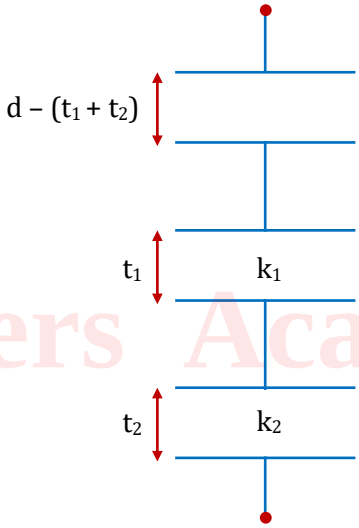


$$V = \frac{\sigma}{\varepsilon_0}(d-(t_1+t_2)) + \frac{\sigma}{\varepsilon_0 K_1}t_1 + \frac{\sigma}{\varepsilon_0 K_2}t_2$$

$$\therefore \quad C = \frac{Q}{V} = \frac{\sigma A}{V} = \frac{\sigma A}{\frac{\sigma}{\varepsilon_0}(d-(t_1+t_2)+\frac{\sigma}{\varepsilon_0 K_1}t_1+\frac{\sigma t_2}{\varepsilon_0 K_2}}$$

$$= \frac{A\varepsilon_0\ K_1K_2}{K_2K_1(d-(t_1+t_2)) + K_2t_1 + K_1t_2}$$

Above question can also be solved by using the concept of series and parallel combination. We can redraw the given configuration as follows.



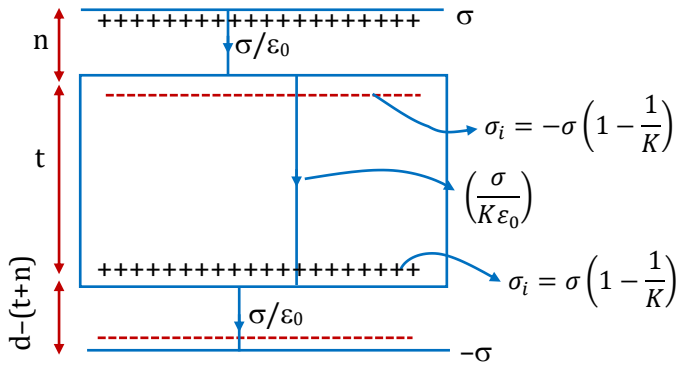
$$C_1 = \frac{A\varepsilon_0}{d-(t_1+t_2)}$$

$$C_2 = \frac{k_1 A\varepsilon_0}{t_1}$$

$$C_3 = \frac{k_2 A\varepsilon_0}{t_2}$$

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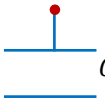
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$



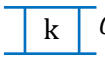
Above question can also be solved by using concept of series and parallel combination.

We can redraw the given configuration as follow.

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$$C_1 = \frac{A\varepsilon_0}{n}$$



$$C_2 = \frac{kA\varepsilon_0}{t}$$

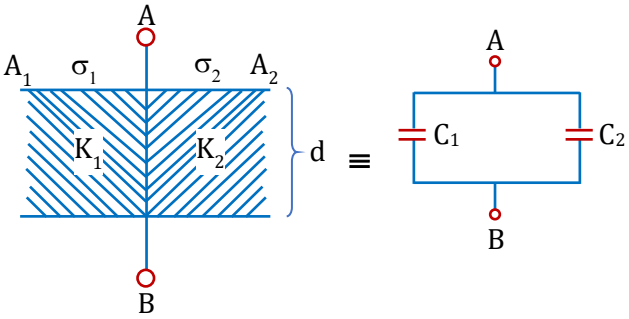


$$C_3 = \frac{A\varepsilon_0}{d-(t+n)}$$

$$\frac{1}{C_{eq.}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

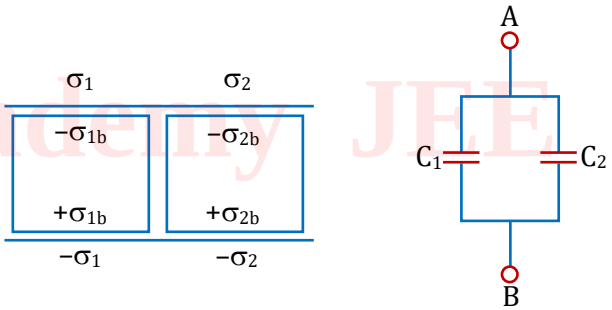
Multiple dielectrics completely fill the capacitor

Lets try to calculate capacitance between A and B if two dielectric slabs of dielectric constant K_1 and K_2 of area A_1 and A_2 and each of thickness d are inserted between the plates of parallel plate capacitor of plate area A as shown in figure. ($A_1 + A_2 = A$)



$$C_1 = \frac{A_1K_1\in_0}{d}, C_2 = \frac{A_2K_2\in_0}{d}$$

As dielectrics are between same metal plates therefore potential difference and hence electric field will be same either calculated using electric field in any of dielectric region. Therefore, if dielectric constant of these two dielectric is different then charges on plate will be different in two regions.



$$E_1 = \frac{V}{d} = \frac{\sigma_1}{K_1\in_0}, \quad E_2 = \frac{V}{d} = \frac{\sigma_2}{K_2\in_0}$$

$$\sigma_1 = \frac{K_1\in_0 V}{d}, \sigma_2 = \frac{K_2\in_0 V}{d}$$

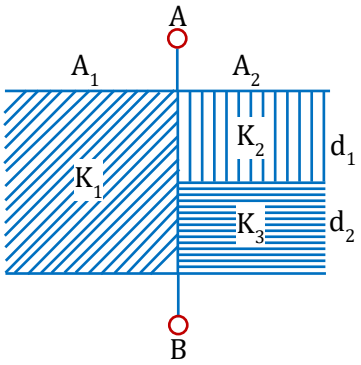
$$C = \frac{Q_1+Q_2}{V} = \frac{\sigma_1A_1+\sigma_2A_2}{V} = \frac{K_1\in_0 A_1}{d} + \frac{K_2\in_0 A_2}{d}$$

The combination is equivalent to : $C = C_1 + C_2$. Hence above question could have been solved by using series and parallel combination.

Illustration 25:

Find out capacitance between A and B if three dielectric slabs of dielectric constant K_1 of area A_1 and thickness d, K_2 of area A_2 and thickness d_1 and K_3 of area A_2 and thickness d_2 are inserted between the plates of parallel plate capacitor of plate area A as shown in figure.

(Given distance between the two plates $d = d_1+d_2$).



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Solution:

It is equivalent to $C = C_1 + \frac{C_2 C_3}{C_2 + C_3}$

$$C = \frac{A_1 K_1 \in_0}{d_1 + d_2} + \frac{\frac{A_2 K_2 \in_0}{d_1} \cdot \frac{A_2 K_3 \in_0}{d_2}}{\frac{A_2 K_2 \in_0}{d_1} + \frac{A_2 K_3 \in_0}{d_2}}$$

$$= \frac{A_1 K_1 \in_0}{d_1 + d_2} + \frac{A_2^2 K_2 K_3 \in_0^2}{A_2 K_2 \in_0 d_2 + A_2 K_3 \in_0 d_1}$$

$$= \frac{A_1 K_1 \in_0}{d_1 + d_2} + \frac{A_2 K_2 K_3 \in_0}{K_2 d_2 + K_3 d_1}$$

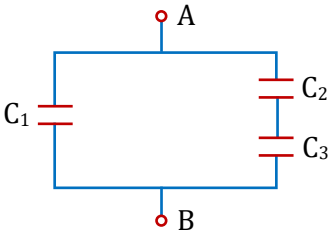
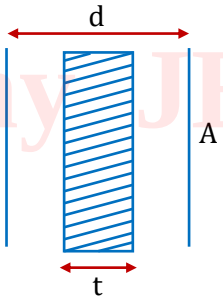


Illustration 26:

A parallel plate isolated condenser consists of two metal plates of area A and separation $'d'$. A slab of thickness $'t'$ and dielectric constant K is inserted between the plates with its faces parallel to the plates and having the same surface area as that of the plates. Find the capacitance of the system. If $K = 2$, for what value of t/d will the capacitance of the system be $3/2$ times that of the condenser with air filling the full space? Calculate the ratio of the energy in the two cases and account for the energy change (assuming q charge on the plate to be constant).

Solution:

$$C' = \frac{A \in_o}{\frac{t}{k} + d - t}$$



(i) Without dielectric, $C = \frac{A \in_o}{d}$

With dielectric, $C' = \frac{A \in_o}{\frac{t}{2} + d - t} = \frac{3C}{2} = \frac{3}{2}\frac{A \in_o}{d} \Rightarrow \frac{d}{t} = \frac{3}{2}$

(ii) Energy $= \frac{Q^2}{2C}$

Energy in 1st case, $E_1 = \frac{q^2}{2C}$

Energy is 2nd case

$E_2 = \frac{q^2}{2C'}$

$\frac{E_1}{E_2} = \frac{C'}{C} = \frac{3}{2}$

(iii) $\Delta E = E_2 - E_1 = \frac{q^2}{2C'} - \frac{q^2}{2C} = \frac{q^2}{2}\left[\frac{2}{3C} - \frac{1}{C}\right] = -\frac{q^2}{6C} = -\frac{q^2 d}{6A\in_0}$

Illustration 27:

The distance between the plates of a parallel-plate capacitor is 0.05 m. A field of 3×10^4 V/m is established between the plates of capacitor by connecting with battery. Now capacitor is disconnected from the battery and an uncharged metal plate of thickness 0.01 m is inserted between the plates of capacitor. Calculate new potential difference between the plates of capacitor. What would be the potential difference if a plate of same thickness and dielectric constant $K = 2$ is introduced in place of metal plate ?

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Solution:

(i) In case of a capacitor as $E = (V/d)$, the potential difference between the plates before the introduction of metal plate

$V = E \times d = 3 \times 10^4 \times 0.05 = 1.5\ kV$

(ii) Now as after charging battery is removed, capacitor is isolated so q = constant. If C' and V' are the capacity and potential after the introduction of plate, $q = CV = C'V'$ i.e., $V' = \frac{C}{C'}V$

And as $C = \frac{\varepsilon_0 A}{d}$ and $C' = \frac{\varepsilon_0 A}{(d-t)+(t/K)} = V' = \frac{(d-t)+(t/K)}{d} \times V$

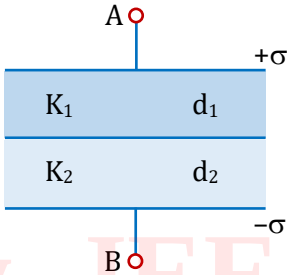
So, in case of metal plate as $K = \infty$, $V_M = \left[\frac{d-t}{d}\right] \times V = \left[\frac{0.05-0.01}{0.05}\right] \times 1.5 = 1.2$ kV

And if instead of metal plate, dielectric with $K = 2$ is introduced

$$V_D = \left[\frac{(0.05-0.01)+\left(\frac{0.01}{2}\right)}{0.05}\right] \times 1.5 = 1.35\ kV$$

Illustration 28:

Find out capacitance between A and B if two dielectric slabs of dielectric constant K_1 and K_2 of thickness d_1 and d_2 and each of area A are inserted between the plates of parallel plate capacitor of plate area A as shown in figure.



Solution:

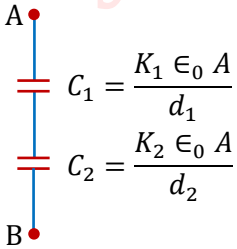
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$$C = \frac{\sigma A}{V} \quad \left(C = \frac{Q}{V}\ ;\ q = \sigma A\right)$$

$$V = E_1 d_1 + E_2 d_2 = \frac{\sigma d_1}{K_1 \in_0} + \frac{\sigma d_2}{K_2 \in_0} = \frac{\sigma}{\in_0}\left(\frac{d_1}{k_1} + \frac{d_2}{k_2}\right)$$

$\therefore$ $C = \frac{A\in_0}{\frac{d_1}{K_1}+\frac{d_2}{K_2}}$

$\Rightarrow$ $\frac{1}{C} = \frac{d_1}{AK_1\in_0} + \frac{d_2}{AK_2\in_0}$



This formula suggests that the system between A and B can be considered as series combination of two capacitors.

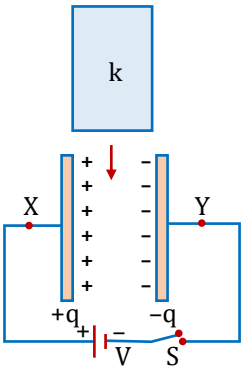
Comparison of E (Electric Field), σ (Surface Charges Density) and Other Parameters:

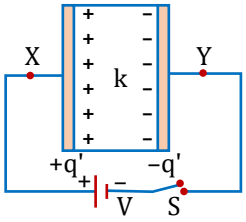
Comparison of E (electric field), σ (surface charge density), Q (charge), C (capacitance) before and after inserting a dielectric slab between the plates of a parallel plate capacitor

Case 1: When constant potential difference across capacitor plates is maintained.

- Parallel plate capacitor of capacitance C connected to a voltage source V due to which the capacitor is charged to the steady state charge $q = CV$. As battery remain connected during insertion of slab the potential difference across capacitor does not change in this case.

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(a) Initial Capacitance, $C = \frac{\epsilon_0 A}{d}$

(b) Initial, $Q = CV$

(c) Initial electric field, $E = \frac{\sigma}{\epsilon_0} = \frac{CV}{A\epsilon_0}$

(d) Initially potential difference between the plates

$Ed = V$

$E = \frac{V}{d}$

$\frac{V}{d} = \frac{\sigma}{\epsilon_0}\ \frac{V}{d} = \frac{\sigma'}{K\epsilon_0}$

Equating both

$\frac{\sigma}{\epsilon_0} = \frac{\sigma'}{K\epsilon_0}$

$\sigma' = K\sigma$

(a) Final capacitance, $C' = \frac{A\epsilon_0 K}{d}$

(b) $Q' = C'V$

(c) $E' = \frac{\sigma'}{K\epsilon_0} = \frac{CV}{A\epsilon_0}$

(d) Here potential difference between the plates

$E'd = V$

$E' = \frac{V}{d}$

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Or we can say that after inserting dielectric final charge becomes KCV which is K times the initial charge. Hence $CV(K-1)$ charge has flown through battery does a work of $CV^2(K\text{-}1)$ in the insertion process.

(e) Energy Stored in capacitor = $CV^2/2$

Energy equation for the capacitor circuit in dielectric insertion process will be

$W_{batt} + W_{ext} = \Delta U$ + Heat.

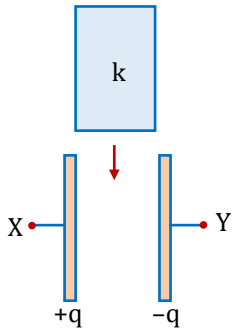
If dielectric is inserted slowly and circuit has negligible resistance then heat loss will be negligible and some negative work will be done by external force.

(e) Energy Stored in Capacitor = $KCV^2/2$

Case 2: When plates of capacitor have constant charge:

- Parallel plate capacitor of capacitance C which is charged to a charge q and disconnected from the voltage source. When the dielectric slab is inserted in this capacitor which fills the space between plates then after insertion of dielectric the physical quantities associated with capacitor changes as listed in table below. As capacitor is charged and not connected in any circuit, its charge will remain constant and does not change while insertion of dielectric slab.

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Parameter before insertion of dielectric.

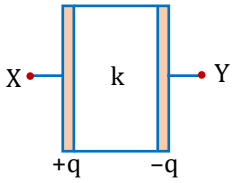
(a) Initial Capacitance $C = \frac{\epsilon_0 A}{d}$

(b) Initial charge = Q

(c) Initial electric field = E

(d) Initial voltage = V

(e) Initial energy stored = U



Parameter after insertion of dielectric

(a) Final Capacitance = kC

(b) Final charge = Q

(c) Final electric field = E/k

(d) Final voltage = V/k

(e) Final energy stored = U/k

Illustration 29:

The parallel plates of a capacitor have an area 0.2 m^2 and are 10^{-2} m apart. The original potential difference between them is 3000 V, and it decreases to 1000 V when a sheet of dielectric is inserted between the plates filling the full space. Compute: ($\epsilon_0 = 9 \times 10^{-12}$ S. I. units)

(i) Original capacitance C_0.

(ii) The charge Q on each plate.

(iii) Capacitance C after insertion of the dielectric.

(iv) Dielectric constant K.

(v) Permittivity ϵ of the dielectric.

(vi) The original field E_0 between the plates.

(vii) The electric field E after insertion of the dielectric.

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Solution:

(i) $C_0 = \frac{\epsilon_0 A}{d} = \frac{0.2\epsilon_0}{10^{-2}} = 20\ \epsilon_0 = 20 \times 9 \times 10^{-12} = 180\ pF$

(ii) $Q = C_0 V = 180 \times 10^{-12} \times 3000 = 5.4 \times 10^{-7}\ C$

(iii) $C_1 = \frac{Q}{V_1} = \frac{5.4 \times 10^{-7}}{1000} = 540\ pF$

(iv) $K = \frac{C_1}{C_0} = \frac{540}{180} = 3$

(v) $\epsilon = \epsilon_r \epsilon_0 = K\epsilon_0 = 3 \times 9 \times 10^{-12} = 27 \times 10^{-12}\ C^2 N^{-1} m^{-2}$

(vi) $E_0 = \frac{V}{d} = \frac{3000}{10^{-2}} = 3 \times 10^5\ V/m$

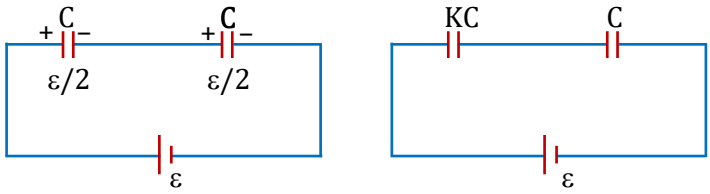
(vii) $E = \frac{V_1}{d} = \frac{1000}{10^{-2}} = 1 \times 10^5\ V/m$

Illustration 30:

Two parallel plate air capacitors each of capacitance C were connected in series to a battery with e.m.f. ε. Then one of the capacitors was filled up with uniform dielectric of relative permittivity k. How many times did the electric field strength in that capacitor decrease? What amount of charge flows through the battery?

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Solution:



$$V_1 = \frac{Q_1}{kC} = \frac{\varepsilon}{k+1}, E_1 = \frac{V_1}{d} = \frac{1}{k+1}\cdot\frac{\varepsilon}{d} \text{ and } E = \frac{Q}{dC} = \frac{\varepsilon}{2d}$$

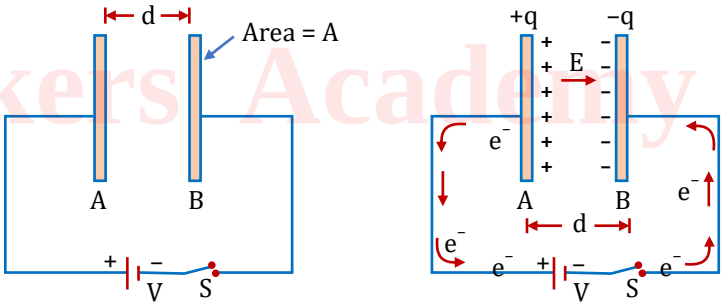
$$E_1 = \frac{2}{(k+1)}E \Rightarrow \frac{1}{2}(1+k) \text{ times decrease}$$

$$\Delta q = Q_1 - Q = \frac{kC\varepsilon}{k+1} - \frac{C\varepsilon}{2} = \frac{C\varepsilon(2k-k-1)}{2(k+1)}$$

$$\Rightarrow \quad \Delta q = \frac{1}{2}\frac{C\varepsilon(k-1)}{(k+1)}$$

Principle

Parallel plate capacitor with plates A and B and a battery of emf V. If we close the switch positive terminal of battery which is connected to the plate A will start pulling electrons from the plate A and will transfer these electrons to the plate B of the capacitor. On plate A positive charge will appear due to loss of electrons and as these electrons are transferred to plate B by battery a negative charge will appear on plate B.



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These charges will reside only on the inner surface of plates A and B and outer surface will not carry any charge because overall charge of the system of this parallel plate capacitor is zero.

Electric Field and Steady State

Charges gets accumulated on the plates and hence gradually electric field is increasing between the plates. As electric field is uniform between the plates and plate separation is d hence potential difference between the plates $(V_A - V_B) = Ed$.

This potential difference is also increasing and as soon as this becomes equal to potential difference across the battery no further charge will flow and current will become zero in the circuit. In this situation we can say capacitor is charged having charge of magnitude $q = CV$ and as no further accumulation takes place we can say that capacitor is in steady state.

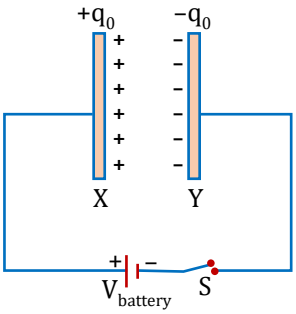
Conclusion

Whenever capacitor is connected across battery of constant potential difference then in steady state charges on capacitor will be $q = CV$ where V is the potential difference applied (may be battery of constant emf) and C is the capacitance of capacitor.

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Charging of Capacitor when Capacitor has initially some charge

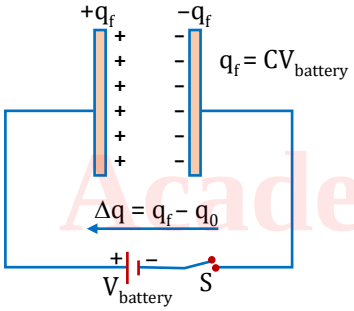
If capacitor having initial charge on its plates as q_0. In this state of initial charge on capacitor if switch is closed then battery will modify the charges on capacitor plates which will depend upon the initial charge on it.



Due to initial charge on capacitor q_0 the initial potential difference of the capacitor is $V_0 = q_0/C$.

Case 1: If $V_0 < V_{battery}$

Then on closing the switch battery will supply charges to capacitor and final charge on capacitor will increase until the potential difference across capacitor becomes equal to that of battery.

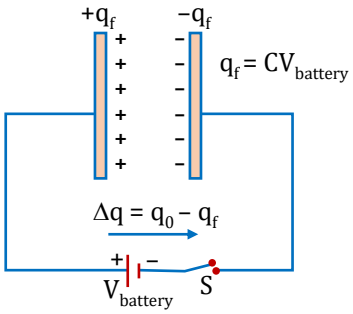


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If $\frac{q_0}{C} < V_{battery} \quad \Rightarrow q_f > q_0$

Thus, the final charge on capacitor plates will be given as $q_f = CV_f$.

Case 2: If $V_0 > V_{battery}$



Then on closing the switch charge will flow into the battery from the capacitor and final charge on capacitor will decrease until the potential difference across capacitor decreases and becomes equal to that of the battery.

If $\frac{q_0}{C} > V_{battery} \quad \Rightarrow q_f < q_0$

The final charge on capacitor in this state will also be given by $q_f = CV_f$

On connecting a capacitor across a battery charges flows from one plate of capacitor to another plate due to battery and as wires are conducting with very low or negligible resistance, charges flow very fast and quickly.

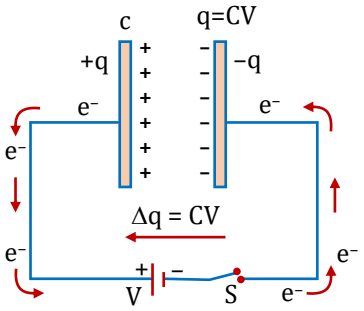
Capacitor plates attain the steady state charges and finally current in circuit becomes zero.

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Heat Loss in Charging

Principle:

In the process free electrons in the conducting wires gain kinetic energy while transferring and finally all charges come to rest thus the kinetic energy attained by these free electrons is finally dissipated in form of heat radiations to the surrounding.



(a) Work by battery (W):

When the switch was open the initial charge on capacitor was zero and there was no energy stored in capacitor. When switch is closed battery supplies a charge on capacitor plates and in the process amount of charge flowing through battery is $q = CV$ thus the work done by battery is given as $W = qV = (CV)V = CV^2$.

(b) Energy Stored in Capacitor (ΔU):

In steady state charge on capacitor plates becomes $q = CV$. Thus, in steady state the potential difference across capacitor plates becomes V and energy stored in it is given by equation $U = \frac{1}{2}CV^2$.

(c) Heat Produced (H):

In the process of charging the capacitor, work done by battery is CV^2 and out of this half is stored in capacitor as electric field remaining energy which the battery supplied as work in charging is dissipated as heat which is given as

$H = W - \Delta U$

$H = CV^2 - CV^2/2 = CV^2/2$

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Illustration 31:

A capacitor having a capacitance of 200 μF is charged to a potential difference of 20V. The charging battery is disconnected and the capacitor is connected to another battery of emf 10V with the positive plate of the capacitor joined with the positive terminal of the battery.

(a) Find the charges on the capacitor before and after the reconnection in steady state.

(b) Find the net charge flown through the 10 V battery

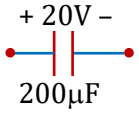
(c) Is work done by the battery or is it done on the battery? Find its magnitude.

(d) Find the decrease in electrostatic field energy.

(e) Find the heat developed during the flow of charge after reconnection.

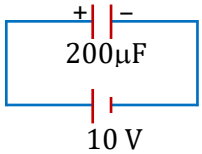
Solution:

(a) Charge on capacitor before connection



$Q_1 = CV_1 = 4000\ \mu C$

Charge on capacitor after connection



$Q_2 = CV_2 = 200 \times 10 = 2000\ \mu C$

(b) Charge flown through the 10V battery = 4000 − 2000 = 2000 μC

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(c) Work is done on the battery.

(d) The decrease in electrostatic field energy.

$= U_i - U_f = \frac{1}{2} CV_1^2 - \frac{1}{2} CV_2^2 = \frac{1}{2} \times 200 \times (20)^2 - \frac{1}{2} \times 200\ (10)^2$

$= 30\ mJ$

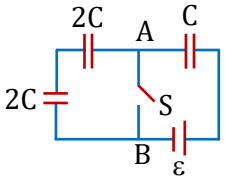
(e) $W = \Delta U + \Delta H$

$-20\ mJ = -30\ mJ + \Delta H$

$[\Delta H = 10\ mJ]$

Illustration 32:

Consider the situation shown in the figure. The switch S is open for a long time and then closed and again steady state reached then



(a) Find the charge flown through the battery after the switch S is closed.

(b) Find the charge flown through the switch S from B to A.

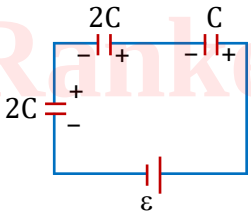
(c) Find the work done by the battery after the switch S is closed.

(d) Find the change in energy stored in the system of capacitors.

(e) Find the heat developed in the system after the switch S is closed.

Solution:

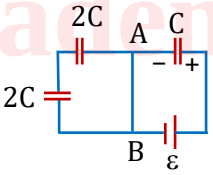
Situation when S is open



$C_{eq.} = C/2$

Charge supplied $Q = C_{eq.}\varepsilon = \frac{C\varepsilon}{2}$

Situation when S is closed



$C_{eq.} = C$

Charge on capacitor $C = C\varepsilon$

After S is closed, voltage across $2C$ capacitor become zero, so charge on it also become zero.

(a) So, charge flown through the battery when the switch S is closed $= C\varepsilon - \frac{C\varepsilon}{2} = \frac{C\varepsilon}{2}$

(b) The charge flown through the switch S is $\frac{C\varepsilon}{2} + \frac{C\varepsilon}{2} = C\varepsilon$ (A to B) $= -C\varepsilon$ (B to A)

(c) Work done by the battery $= Q_{\text{supplied}}\ V = \frac{C\varepsilon}{2}\ \varepsilon = \frac{C\varepsilon^2}{2}$

(d) Change in energy stored in the capacitors = energy stored in 2nd case – energy stored in 1st case

$= \frac{1}{2} C_{eq2}\ V^2 - \frac{1}{2} C_{eq1}\ V^2 = \frac{1}{2} \cdot \left(C - \frac{C}{2}\right) \varepsilon^2 = \frac{C\varepsilon^2}{4}$

(e) Heat developed in the system = Work done by battery – change in energy in the capacitors

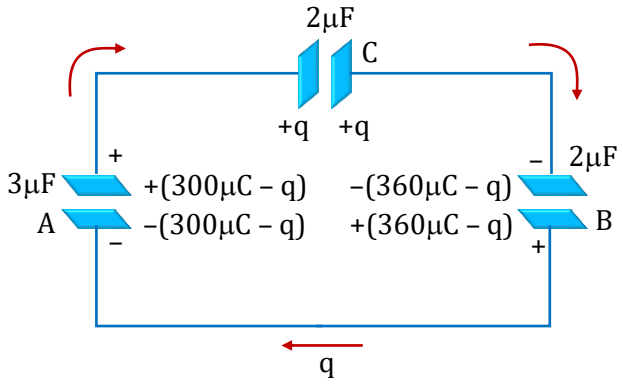
$\frac{C\varepsilon^2}{2} - \frac{C\varepsilon^2}{4} = \frac{C\varepsilon^2}{4}$

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KVL or Charge Flow Method in Capacitor circuit with R = 0

In this method we distribute charge *s* at various capacitors of circuit and then we write potential equation for different loops of circuit.



Analysis in circuit 1

This method is explained by using an illustration in which there are two capacitors of capacitances $4\mu F$ and $8\mu F$ initially charged at $20V$ and $5V$ respectively and connected in a single loop with two batteries of $10V$ and $20V$ along with a switch S.

Here, we will try to find the steady state charges of the two capacitors when switch is closed.

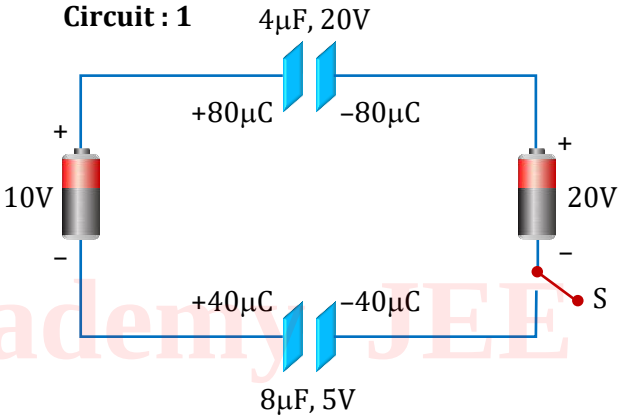
On closing the switch redistribution of charges will take place and a charge q flows through the circuit in anticlockwise direction until new steady state is achieved and the final charges on capacitor plates will be as shown in figure.

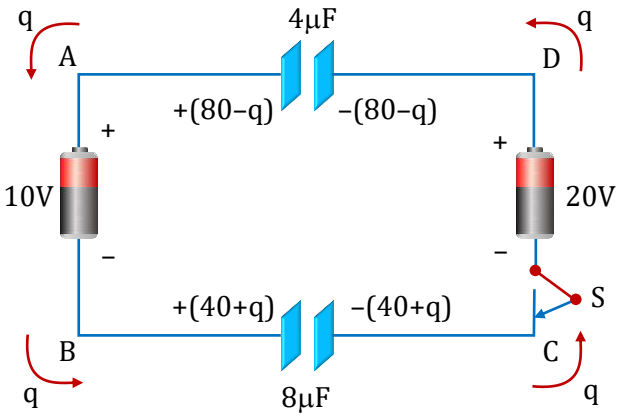
We can write equation of potential for the circuit loop starting from any one point in the loop move either clockwise or anticlockwise and come back to the same point as described.

$$V_A - 10 - \frac{40+q}{8} + 20 + \frac{80-q}{4} = V_A$$

Solving we get $q = \frac{200}{3}\ \mu C$

Circuit : 1



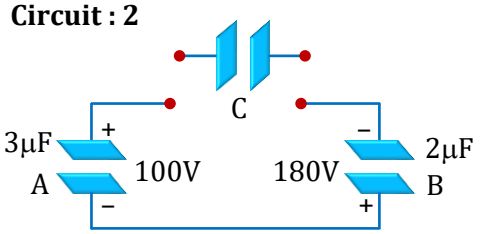


Analysis in circuit 2

Applying *KVL* in another example.

Two capacitors *A* and *B* with capacitances $3\mu F$ and $2\mu F$ are charged to potential difference of $100V$ and $180V$ respectively. The plates of the capacitors are connected as shown with one wire from each capacitor free.

Circuit : 2



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The upper plate of A is positive and that of B is negative. An uncharged $2\mu F$ capacitor C with lead wires falls on the free ends to complete the circuit. We will try to calculate the final charge on the three capacitors and the amount of electrostatic energy stored in the system before and after the capacitor C falls and the circuit is closed.

Let q be the charge flowing in clockwise sense during redistribution of charge as shown. We can write the potential equation for this loop as

$$-\frac{q}{2}+\frac{360-q}{2}+\frac{300-q}{3}=0$$

$$q = 210\ \mu C$$

Final charges on the capacitors are

$$q_A = 300 - 210 = 90\ \mu C$$

$$q_B = 360 - 210 = 150\ \mu C$$

$$q_C = 210 = 210\ \mu C$$

Initial electrostatic energy stored is given as

$$U_i = \frac{1}{2}\times(3\times10^{-6})(100)^2+\frac{1}{2}\times(2\times10^{-6})(180)^2$$

$$U_i = 4.74\times10^{-2}J = 47.4\times10^{-3}J = 47.4mJ$$

Final electrostatics energy stored is given as

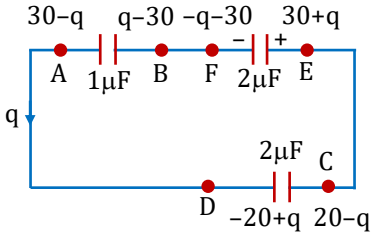
$$U_f = \frac{(90\times10^{-6})^2}{2\times(3\times10^{-6})}+\frac{(150\times10^{-6})^2}{2\times(2\times10^{-6})}+\frac{(210\times10^{-6})^2}{2\times(2\times10^{-6})}$$

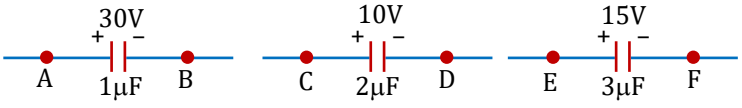
$$U_f = 1.8\times10^{-2}J = 18mJ$$

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Illustration 33:

Three capacitors as shown of capacitance $1\mu F$, $2\mu F$ and $2\mu F$ are charged upto potential difference 30 V, 10 V and 15 V respectively. If terminal A is connected with D, C is connected with E and F is connected with B. Then find out charge flow in the circuit and find the final charges on capacitors.





Solution:

Let charge flow is q.

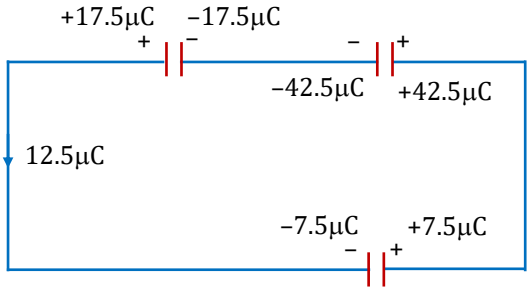
Now applying Kirchhoff's voltage law

$$-\frac{(q-20)}{2}-\frac{(30+q)}{2}+\frac{30-q}{1}=0$$

$$-2q = -25$$

$$q = 12.5\ \mu C$$

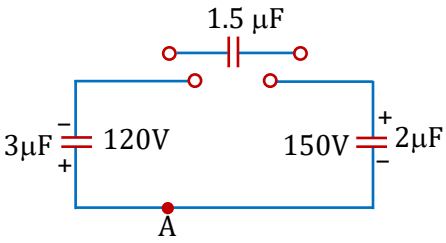
Final charges on plates.



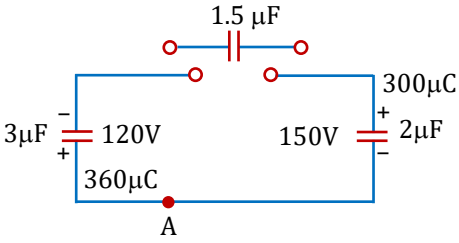
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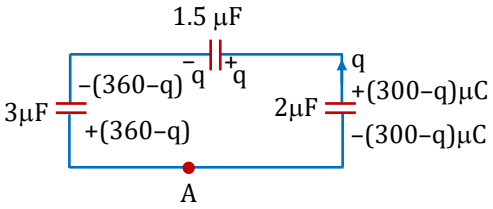
Illustration 34:

Two capacitors of 2 μF and 3 μF are charged to 150 volt and 120 volts respectively. The plates of a capacitor are connected as shown in the fig. A discharged capacitor of capacity 1.5 μF falls to the free ends of the wire and connected through the free ends of the wire, then find the charge on each capacitor at steady state.



Solution:





$$V_A + \frac{(300-q)}{2} - \frac{q}{1.5} + \frac{360-q}{3} = V_A \qquad \text{...(i)}$$

By solving this equation, we get $\Rightarrow q = 180\mu C$

Charge on $1.5\mu F$ capacitor is = $180\mu C$

Charge on $2\mu F$ capacitor is = $300 - 180 = 120\mu C$

Charge on $3\mu F$ capacitor is = $360 - 180 = 180\mu C$

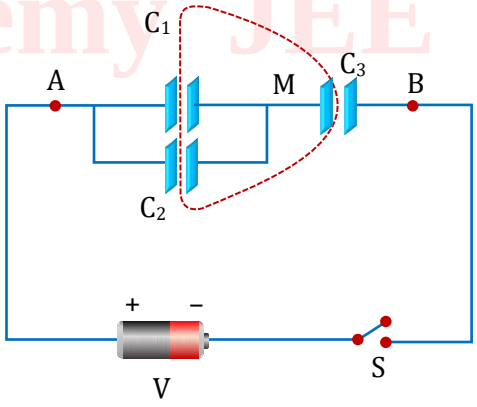
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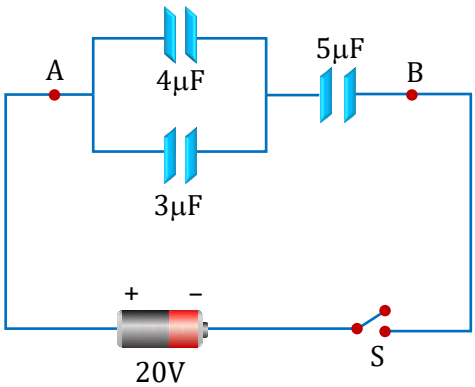
Nodal Analysis

There are some cases in which capacitors are connected in such combinations which are neither series nor parallel called mixed combinations. Such cases can be easily handled by a specific method of solving capacitive circuits called 'Nodal Analysis'.

'Node' in capacitive circuit is an isolated part of circuit at which two or more plates of capacitors are connected. Shows a capacitive circuit in which C_1 and C_2 are connected in parallel and this group is connected in series with C_3 and connected across a battery of voltage V. In this circuit also, a node is shown at junction M by closed dotted line.

We will discuss the understanding and application of nodal analysis using an illustration shown. In this circuit we will calculate the equivalent capacitance of the circuit across terminals AB and steady state charge on each capacitor after closing the switch. There are three steps to be followed for solving this circuit using nodal analysis.



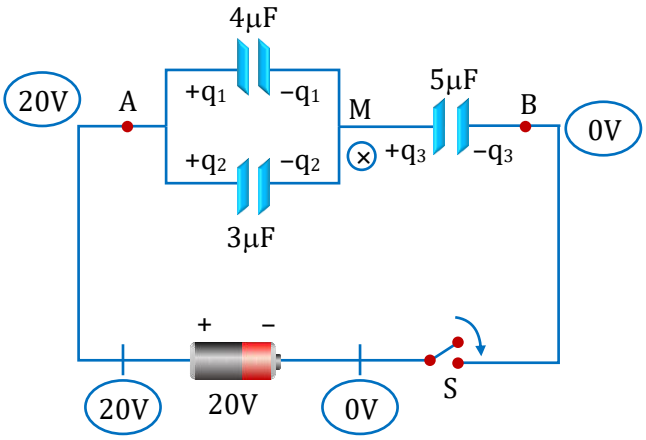


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Step-I : Identifying the Nodes of Circuit

Step-II : Distribution of Potential in Circuit

Distribute potential at every point in the circuit. Starting with the battery terminals and first considering the potential at negative terminal of the battery as zero. At the node junction M as it is neither connected to left or right side of battery directly, we can consider its potential to be x. Starting with zero potential reference from negative terminal of battery and distributing potentials at all other points of circuit. For nodes where potential is unknown, we give variable potentials x, y, z etc. if more than one node are there.



Step-III : Writing Nodal Equations

On a node charges can only be induced so charges on these plates must be conserved. If capacitors are initially uncharged, net charge on a node must remain zero and if these are initially charged then sum of charges must remain same.

$$q_3 - q_1 - q_2 = 0$$

$$q_1 = 4 \times (20 - x)\mu C \quad ...(1)$$

$$q_2 = 3 \times (20 - x)\mu C \quad ...(2)$$

and $$q_3 = 5 \times x\mu C \quad ...(3)$$

Substituting the values of charges in equation (1), we get

$$5x - 4(20 - x) - 3(20 - x) = 0 \quad ...(4)$$

$$\Rightarrow \quad 12x = 140$$

$$\Rightarrow \quad x = \frac{140}{12} = \frac{35}{3}\,V \quad ...(5)$$

$$\Delta q_0 = |q_1 + q_2| = |q_3| = \frac{175}{3}\mu C$$

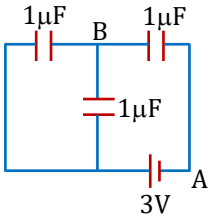
Thus, equivalent capacitance in this case can be given as

$$C_{eq} = \frac{\Delta q_0}{V_{\text{battery}}} = \frac{\left(\frac{175}{3}\right)}{20} = \frac{35}{12}\mu F \quad ...(6)$$

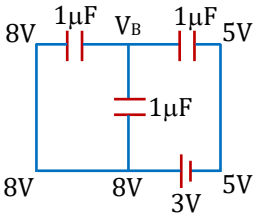
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Illustration 35:

If potential of A is $5V$, then potential of B in volt is



Solution:



$$1(8 - V_B) + 1(8 - V_B) + 1(5 - V_B) = 0$$

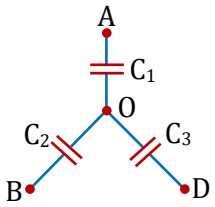
$$21 = 3V_B$$

$$V_B = 7V$$

For More Join: @IITJEE_Advanced

Illustration 36:

Three uncharged capacitors of capacitance $C_1 = 1\mu F$, $C_2 = 2\mu F$ and $C_3 = 3\mu F$ are connected as shown in the figure. The potential of point A, B and D are 10 volt, 25 volt and 20 volt respectively. Determine the potential at point O.



Solution:

$$(V_A - V_o)C_1 + (V_B - V_o)C_2 + (V_D - V_o)C_3 = 0$$

$$V_o = \frac{V_A C_1 + V_B C_2 + V_D C_3}{C_1 + C_2 + C_3}$$

$$= \frac{10\times1+25\times2+20\times3}{1+2+3} = 20\ V$$

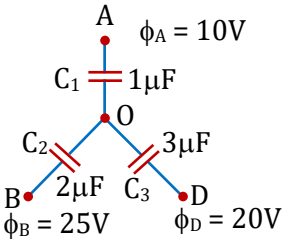
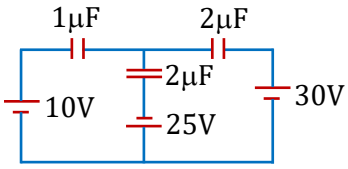
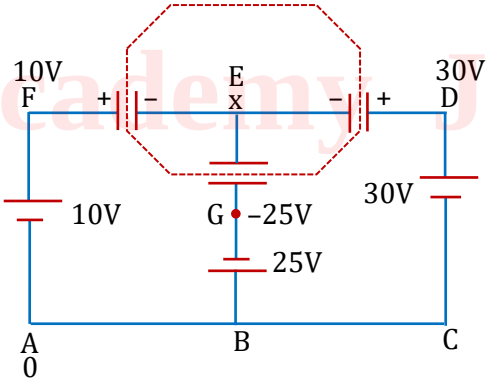


Illustration 37:

In the given circuit find out the charge on each capacitor. (Initially they are uncharged)





Solution:

Let potential at A is 0, so at D it is 30 V, at F it is 10 V and at point G potential is $-25V$ and let potential at E is x. Now apply Kirchhoff's Ist law at point E. (Total charge of all the plates connected to 'E' must be same as before i.e. 0)

$\therefore$ $(x - 10) + (x - 30)2 + (x + 25)2 = 0$

$5x = 20$

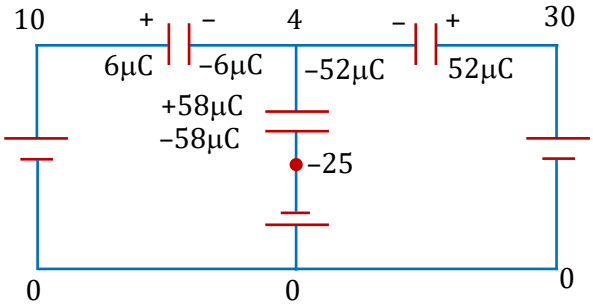
$x = 4\ V$

Final charges:

$Q_{2\mu F} = (30 - 4)2 = 52\ \mu C$

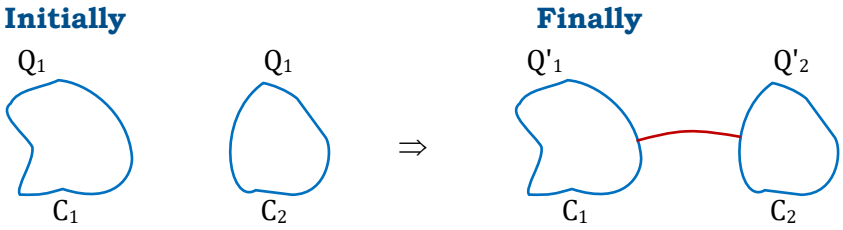
$Q_{1\mu F} = (10 - 4) = 6\mu C$

$Q_{2\mu F} = (4 - (-25))2 = 58\ \mu C$



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Sharing of Charges on Joining two Conductors (By a conducting wire)



Principle:

Whenever there is potential difference, there will be movement of charge. If only electric force act on charge then they have tendency to move from **high potential energy** to **low potential energy**. Therefore, positive charge moves from **high potential** to **low potential** and negative charge moves from **low potential** to **high potential**. This movement of charge will continue till the potential difference between the conductors vanishes.

Formulae related with redistribution of charges:

Before connecting the conductors		
Parameter	**Ist conductor**	**IInd conductor**
Capacitance	C_1	C_2
Charge	Q_1	Q_2
Potential	V_1	V_2

After connecting the conductors		
Parameter	**Ist conductor**	**IInd conductor**
Capacitance	C_1	C_2
Charge	Q_1'	Q_2'
Potential	V	V

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Calculation:

$$V = \frac{Q_1'}{C_1} = \frac{Q_2'}{C_2} \Rightarrow \frac{Q_1'}{Q_2'} = \frac{C_1}{C_2}$$

But $Q_1 + Q_2 = Q'_1 + Q'_2$

$\therefore$ $$V = \frac{Q_1 + Q_2}{C_1 + C_2} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

$\therefore$ $$Q_1' = \frac{C_1}{C_1 + C_2}(Q_1 + Q_2)$$

and $$Q_2' = \frac{C_2}{C_1 + C_2}(Q_1 + Q_2)$$

Heat loss during redistribution:

$$\Delta H = \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V_1 - V_2)^2$$

The loss of energy is in the form of Joule heating in the wire.

Note:

Always put Q_1, Q_2, V_1 and V_2 with sign.

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Illustration 38:

Two conductors having capacities $2\mu F$ and $5\mu F$ and potentials 2 volt and 10 volt respectively. Calculate the ratio of their charges after connecting by a wire.

Solution:

$$\frac{Q'_1}{Q'_2} = \frac{C_1 V_{com}}{C_2 V_{com}} = \frac{2 \times 10^{-6}}{5 \times 10^{-6}} = \frac{2}{5}$$

Illustration 39:

Two charged metal spheres of radii R and $2R$ are temporarily placed in contact and then separated. At the surface of each, which sphere has the greater value for the following:

(a) Charge (b) Charge density (c) Potential (d) Electric field

Solution:

(a) In sharing, charge divides in proportion to capacity so,

$$\frac{q_1}{q_2} = \frac{C_1}{C_2} = \frac{R_1}{R_2} = \frac{R}{2R} = \frac{1}{2}$$

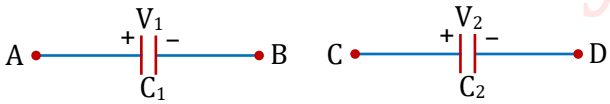
(b) Charge density, $\sigma = \frac{q}{4\pi R^2}$ So, $\frac{\sigma_1}{\sigma_2} = \frac{q_1}{q_2} \times \left[\frac{R_2}{R_1}\right]^2 = \frac{1}{2} \times \left[\frac{2R}{R}\right]^2 = 2$

(c) In case of spherical conductor, $V_S = \frac{q}{4\pi\varepsilon_0 R}$ So, $\frac{V_1}{V_2} = \frac{q_1}{q_2} \times \frac{R_2}{R_1} = \frac{1}{2} \times \frac{2R}{R} = 1$

(d) In case of spherical conductor, $E_S = \frac{q}{4\pi\varepsilon_0 R^2}$ So, $\frac{E_1}{E_2} = \frac{q_1}{q_2} \times \left[\frac{R_2}{R_1}\right]^2 = \frac{1}{2} \times \left[\frac{2R}{R}\right]^2 = 2$

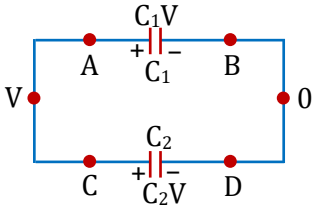
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Sharing of Charge between two Charged Capacitor:

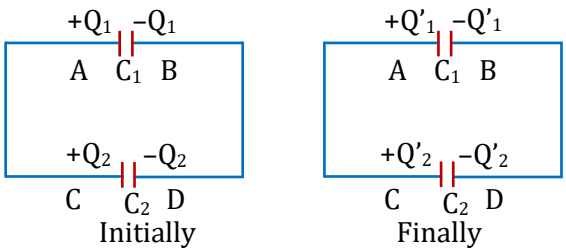


Initially let's say capacitors C_1 is charged by applying potential difference V_1 and similarly capacitor C_2 by applying potential difference V_2.

When two capacitors are C_1 and C_2 are connected as shown in figure. Let potential of B and D is zero and common potential on capacitors is V, then at A and C it will be V.



By charge conservation of plates A and C before and after connection



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Before connecting the conductors		
Parameter	**Ist conductor**	**IInd conductor**
Capacitance	C_1	C_2
Charge	Q_1	Q_2
Potential	V_1	V_2

After connecting the conductors		
Parameter	**Ist conductor**	**IInd conductor**
Capacitance	C_1	C_2
Charge	Q_1'	Q_2'
Potential	V	V

$Q_1 + Q_2 = C_1V + C_2V$ or $C_1V + C_2V = C_1V_1 + C_2V_2$

Which gives, $V = \frac{C_1V_1+C_2V_2}{C_1+C_2}$

This could be understood as $V = \frac{Q_1+Q_2}{C_1+C_2} = \frac{C_1V_1+C_2V_2}{C_1+C_2} = \frac{\text{Total charge}}{\text{Total capacitance}}$

(a) Q_1'(Final charge on capacitor C_1) $= C_1V = \frac{C_1}{C_1+C_2}(Q_1 + Q_2)$

$\Rightarrow$ Q_2'(Final charge on capacitor C_2) $= C_2V = \frac{C_2}{C_1+C_2}(Q_1 + Q_2)$

(b) Heat loss during redistribution:

$$\Rightarrow \Delta H = \frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2 - \frac{1}{2}(C_1 + C_2)V^2$$

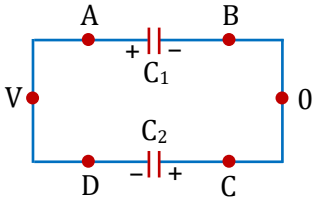
$$= \frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2 - \frac{1}{2}\frac{(C_1V_1 + C_2V_2)^2}{(C_1 + C_2)}$$

$$= \frac{1}{2}\left[\frac{C_1^2V_1^2 + C_1C_2V_1^2 + C_2C_1V_2^2 + C_2^2V_2^2 - C_1^2V_1^2 - C_2^2V_2^2 - 2C_1C_2V_1V_2}{C_1 + C_2}\right]$$

$$= \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V_1 - V_2)^2$$

$$\Delta H = U_i - U_f = \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V_1 - V_2)^2$$

The loss of energy is in the form of Joule heating in the wire.



When oppositely charge terminals are connected then

$\therefore$ $C_1V + C_2V = C_1V_1 - C_2V_2$

$V = \frac{C_1V_1 - C_2V_2}{C_1 + C_2}$ and $H = \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V_1 + V_2)^2$

These two following formula can be remembered for both the cases using proper sign convention.

$$V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

$$H = \frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2 - \frac{1}{2}(C_1 + C_2)V^2$$

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Note:

(i) When plates of similar charges are connected with each other (+ with + and – with –) then put all values (Q_1, Q_2, V_1, V_2) with positive sign.

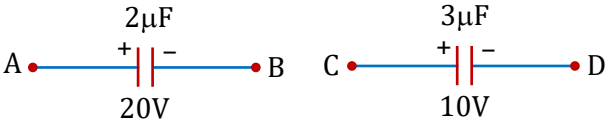
(ii) When plates of opposite polarity are connected with each other (+ with –) then take charge and potential of one of the plate to be negative.

Illustration 40:

Find out the following if A is connected with C and B is connected with D.

(i) How much charge flows in the circuit.

(ii) How much heat is produced in the circuit.



Solution:

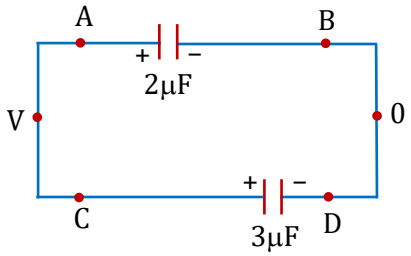
(i) Let potential of B and D is zero and common potential on capacitors is V, then at A and C it will be V.

By charge conservation,

$3V + 2V = 40 + 30$

$5V = 70$

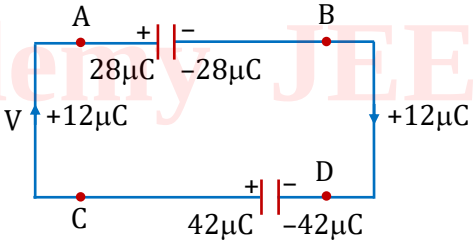
$V = 14$ volt



Charge flow

$= 40 - 28$

$= 12\ \mu C$



Now final charges on each plate is shown in the figure.

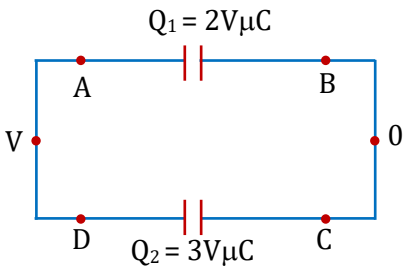
(ii) Heat produced $= \frac{1}{2} \times 2 \times (20)^2 + \frac{1}{2} \times 3 \times (10)^2 - \frac{1}{2} \times 5 \times (14)^2$

$= 400 + 150 - 490$

$= 550 - 490 = 60\ \mu J$

Illustration 41:

Repeat above question if A is connected with D and B is connected with C.



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Solution:

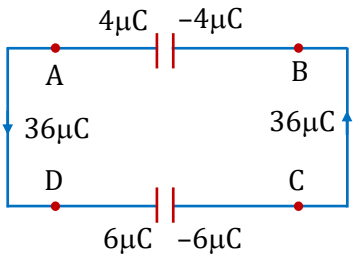
Let potential of B and C is zero and common potential on capacitors is V, then at A and D it will be V

$(2V + 3V)\ \mu C = 40 - 30 = 10\ (\mu C)$

$\Rightarrow \quad V = 2\ volt$

$Q_1 = 2V = 4\mu C$

$Q_2 = 3V = 6\mu C$



Now charge on each plate is shown in the figure

$$\text{Heat produced} = 400 + 150 - \frac{1}{2} \times 5 \times 4$$

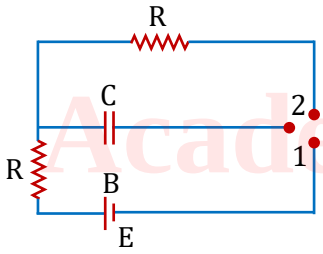
$$= 550 - 10$$

$$= 540\ \mu J$$

Charging of A Capacitor

Charging of a condenser:

(i) In the following circuit. If key 1 is closed then the condenser gets charged. Finite time is taken in the charging process. The quantity of charge at any instant of time t is given by $q = q_0[1 - e^{-(t/RC)}]$



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Where q_0 = maximum final value of charge at $t = \infty$.

According to these equations the quantity of charge on the condenser increases exponentially with increase of time.

(ii) If $t = RC = \tau$ then

$q = q_0\ [1 - e^{-(RC/RC)}] = q_0 \left[1 - \frac{1}{e}\right]$

or $q = q_0\ (1 - 0.37) = 0.63\ q_0 = 63\%$ of q_0

(iii) Time $t = RC$ is known as time constant.

i.e. the time constant is that time during which the charge rises on the condenser plates to 63% of its maximum value.

(iv) The potential difference across the condenser plates at any instant of time is given by

$V = V_0[1 - e^{-(t/RC)}]\ volt$

(v) The potential curve is also similar to that of charge. During charging process an electric current flows in the circuit for a small interval of time which is known as the transient current. The value of this current at any instant of time is given by

$I = I_0[e^{-(t/RC)}]$ ampere

According to this equation the current falls in the circuit exponentially (Fig.).

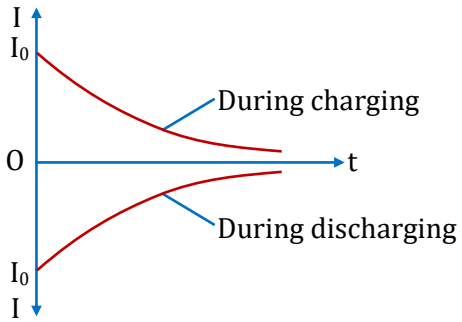
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(vi) If $t = RC = \tau$ = Time constant

$$I = I_0 e^{(-RC/RC)} = \frac{I_0}{e} = 0.37\, I_0$$

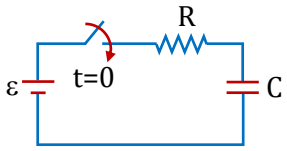
= 37% of I_0

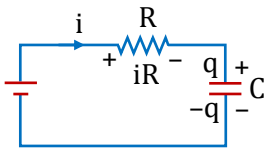
i.e. time constant is that time during which current in the circuit falls to 37% of its maximum value.

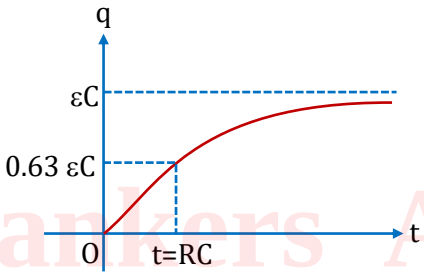


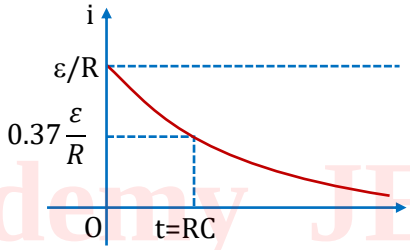
Derivation of formulae for charging of capacitor

It is given that initially capacitor is uncharged. Let at any time charge on capacitor is q. Applying Kirchoff voltage law









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$$\varepsilon - iR - \frac{q}{C} = 0 \Rightarrow iR = \frac{\varepsilon C - q}{C}$$

$$i = \frac{\varepsilon C - q}{CR} \Rightarrow \frac{dq}{dt} = \frac{\varepsilon C - q}{CR}$$

$$\frac{CR}{\varepsilon C - q} . dq = dt$$

$$\int_0^q \frac{dq}{\varepsilon C - q} = \int_0^t \frac{dt}{RC} \Rightarrow -ln\,(\varepsilon C - q) + ln\,\varepsilon C = \frac{t}{RC}$$

$$\ln \frac{\varepsilon C}{\varepsilon C - q} = \frac{t}{RC}\ ;\ \varepsilon C - q = \varepsilon C\, . e^{-t/RC}$$

$$q = \varepsilon C(1 - e^{-t/RC})$$

RC = time constant of the RC series circuit.

After one time constant

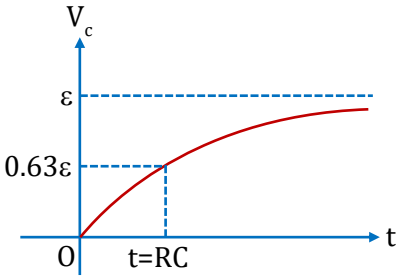
$$q = \varepsilon C\left(1 - \frac{1}{e}\right) = \varepsilon C\,(1 - 0.37) = 0.63\,\varepsilon C$$

Current at any time t

$$i = \frac{dq}{dt} = \varepsilon C\left(-e^{-t/RC}\left(-\frac{1}{RC}\right)\right) = \frac{\varepsilon}{R} e^{-t/RC}$$

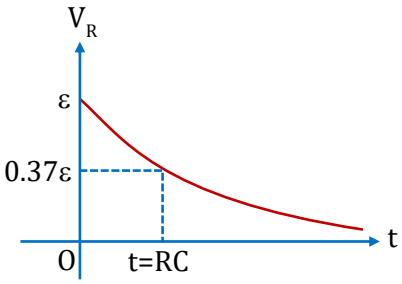
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Voltage across capacitor after one-time constant $V = 0.63\ \varepsilon$



$Q = CV$; $V_C = \varepsilon(1 - e^{-t/RC})$

Voltage across the resistor



$V_R = iR = \varepsilon e^{-t/RC}$

By energy conservation,

Heat dissipated = work done by battery – $\Delta U_{\text{capacitor}}$

$= C\varepsilon(\varepsilon) - (\frac{1}{2}C\varepsilon^2 - 0) = \frac{1}{2}C\varepsilon^2$

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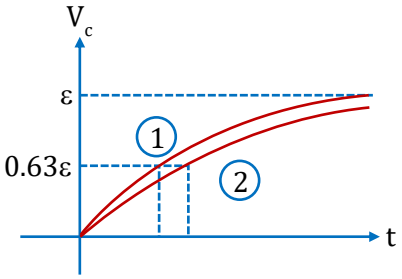
Alternatively:

$$Heat = H = \int_0^\infty i^2 R dt$$

$$= \int_0^\infty \frac{\varepsilon^2}{R^2} e^{-\frac{2t}{RC}} R\, dt = \frac{\varepsilon^2}{R}\int_0^\infty e^{-2t/RC}\, dt = \frac{\varepsilon^2}{R}\left[\frac{e^{-\frac{2t}{RC}}}{-2/RC}\right]_0^\infty$$

$$= -\frac{\varepsilon^2 RC}{2R}\left[e^{-\frac{2t}{RC}}\right]_0^\infty = \frac{\varepsilon^2 C}{2}$$

Note:



In the figure time constant of (2) is more than (1).

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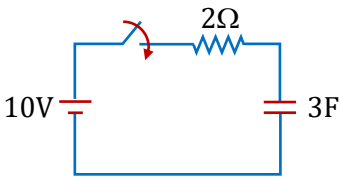
Charging of RC circuit

Illustration 42:

Find out current in the circuit and charge on capacitor which is initially uncharged in the following situations.

(a) Just after the switch is closed.

(b) After a long time when switch was closed.

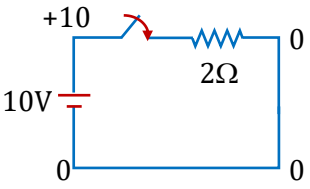


Solution:

(a) For just after closing the switch:

potential difference across capacitor = 0

$\therefore \quad Q_C = 0 \qquad \therefore \quad i = 5A$



(b) After a long time

at steady state current, $i = 0$

and potential difference across capacitor = 10 V

$\therefore \quad Q_C = 3 \times 10 = 30\ C$

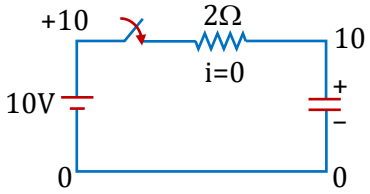
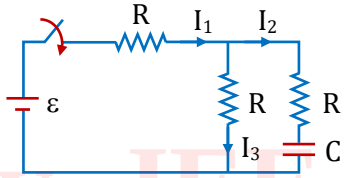


Illustration 43:

Find out current I_1, I_2, I_3, charge on capacitor and $\frac{dQ}{dt}$ of capacitor in the circuit which is initially uncharged in the following situations.



(a) Just after the switch is closed

(b) After a long time when switch is closed.

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Solution:

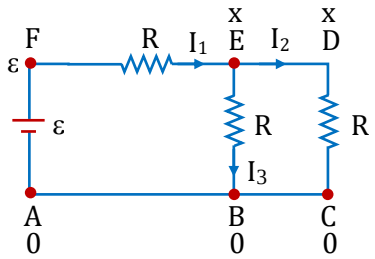
(a) Initially the capacitor is uncharged so its behaviour is like a conductor

Let potential at A is zero so at B and C also zero and at F it is ε. Let potential at E is x so at D also x.

Apply Kirchhoff's I[st] law at point E :

$$\frac{x-\varepsilon}{R}+\frac{x-0}{R}+\frac{x-0}{R}=0 \Rightarrow \frac{3x}{R}=\frac{\varepsilon}{R}$$

$$x=\frac{\varepsilon}{3};\ Q_c=0$$



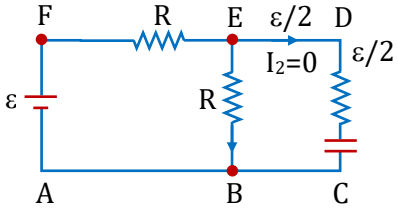
$$\therefore \quad I_1=\frac{-\varepsilon/3+\varepsilon}{R}=\frac{2\varepsilon}{3R},\ I_2=\frac{dQ}{dt}=\frac{\varepsilon}{3R} \text{ and } I_3=\frac{\varepsilon}{3R}$$

Alternatively:

$$i_1=\frac{\varepsilon}{R_{eq}}=\frac{\varepsilon}{R+\frac{R}{2}}=\frac{2\varepsilon}{3R},\ i_2=i_3=\frac{i_1}{2}=\frac{\varepsilon}{3R} \text{ and } \frac{dQ}{dt}=i_2=\frac{\varepsilon}{3R}$$

(b) At $t = \infty$ (finally)

Capacitor completely charged so their will be no current through it.



$$I_2 = 0, \quad I_1 = I_3 = \frac{\varepsilon}{2R}$$

$$V_E - V_B = V_D - V_C = (\varepsilon/2R)R = \varepsilon/2 \Rightarrow Q_C = \frac{\varepsilon C}{2}, \frac{dQ}{dt} = I_2 = 0$$

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	I_1	I_2	I_3	Q	dQ/dt
Time $t = 0$	$\frac{2\varepsilon}{3R}$	$\frac{\varepsilon}{3R}$	$\frac{\varepsilon}{3R}$	0	$\frac{\varepsilon}{3R}$
Finally $t = \infty$	$\frac{\varepsilon}{2R}$	0	$\frac{\varepsilon}{2R}$	$\frac{\varepsilon C}{2}$	0

Illustration 44:

A capacitor is connected to a 36 V battery through a resistance of 20Ω. It is found that the potential difference across the capacitor rises to 12.0 V in 2μs. Find the capacitance of the capacitor.

Solution:

The charge on the capacitor during charging is given by $Q = Q_0(1 - e^{-t/RC})$.

Hence, the potential difference across the capacitor is $V = Q/C = Q_0/C\ (1 - e^{-t/RC})$.

Here, at $t = 2\ \mu s$, the potential difference is $12V$ whereas the steady potential difference is

$Q_0/C = 36V$. *So,* $12V = 36V(1 - e^{-t/RC})$

or $\quad 1 - e^{-t/RC} = \frac{1}{3} \Rightarrow e^{-t/RC} = \frac{2}{3}$

or $\quad \frac{t}{RC} = \ell n\left(\frac{3}{2}\right) = 0.405 \Rightarrow RC = \frac{t}{0.405} = \frac{2\ \mu s}{0.405} \Rightarrow RC = 4.936\ \mu s$

or, $\quad C = \frac{4.936\mu s}{20\Omega} = 0.25\ \mu F.$

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Illustration 45:

A capacitor of capacity 1μF is connected in a closed series circuit with a resistance of 10^7 ohms, an open key and a cell of 2 V with negligible internal resistance:

(i) When the key is switched on at time $t = 0$, find;

(a) The time constant for the circuit.

(b) The charge on the capacitor at steady state.

(c) Time taken to deposit charge equal to half of charge that will deposit at steady state.

(ii) If after completely charging the capacitor, the cell is shorted by zero resistance at time $t = 0$, find the charge on the capacitor at $t = 50$ *s.*

(Given: $e^{-5} = 6.73 \times 10^{-3}$, $ln2 = 0.693$)

Solution:

(i) (a) Time constant $\tau = RC = 10^7 \times 1 \times 10^{-6}$

$= 10$ *sec.*

(b) $Q_0 = CV = 1 \times 10^{-6} \times 2 = 2\ \mu C$

(c) $Q = \frac{Q_0}{2} = Q_0(1 - e^{-t/Rc})$

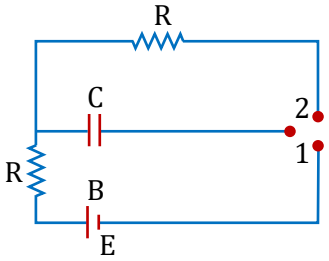
$\frac{t}{RC} = \ell n2 \quad \Rightarrow \quad t = 10\ \ell n2 = 6.93$ *sec.*

(ii) $q = q_0 e^{-t/RC} = 2 \times 10^{-6}\ e^{-50/10} = 2 \times 10^{-6}\ e^{-5} = 1.348 \times 10^{-8}\ C.$

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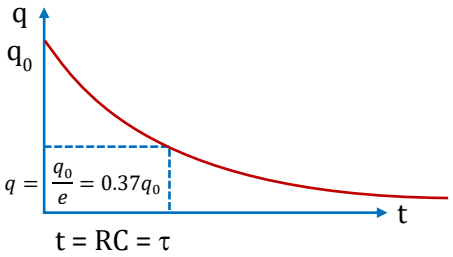
Discharging of a condenser

(i) In the circuit shown below, if key 1 is opened and key 2 is closed then the condenser gets discharged.



(Initial charge on capacitor = q_0)

(ii) The quantity of charge on the condenser at any instant of time t is given by $q = q_0\, e^{-(t/RC)}$



i.e. the charge falls exponentially.

Here q_0 = initial charge of capacitor

(iii) If $t = RC = \tau$ = time constant, then $q = \frac{q_0}{e} = 0.37q_0 = 37\%$ of q_0

i.e., the time constant is that time during which the charge on condenser plates in discharge process, falls to 37%.

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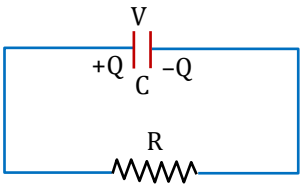
(iv) The dimensions of RC are those of time i.e. $[M^0L^0T^1]$ and the dimensions of $\frac{1}{RC}$ are those of frequency i.e. $[M^0L^0T^{-1}]$.

(v) The potential difference across the condenser plates at any instant of time t is given by $V = V_0e^{-(t/RC)}$ Volt.

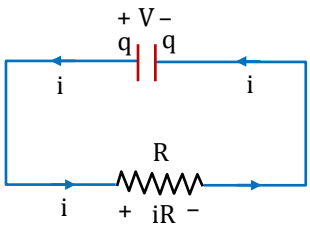
(vi) The transient current at any instant of time is given by $I = -I_0e^{-(t/RC)}$ ampere.

i.e. the current in the circuit decreases exponentially but its direction is opposite to that of charging current. (–ve only means that direction of current is opposite to that at charging current)

Derivation of equation of discharging circuit



Initially



At any time t

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Applying K.V.L.

$$+\frac{q}{C} - iR = 0$$

$$i = \frac{q}{CR}, -\frac{dq}{dt} = \frac{q}{CR}$$

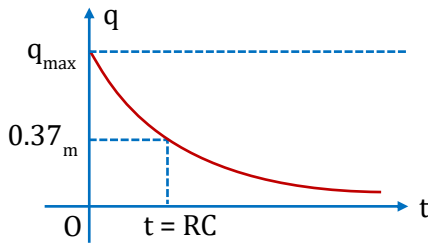
$$\int_Q^q \frac{-dq}{q} = \int_0^t \frac{dt}{CR}$$

$$-\ln\frac{q}{Q} = +\frac{t}{RC}$$

$$q = Q.\, e^{-t/RC}$$

$$i = -\frac{dq}{dt} = \frac{Q}{RC}\, e^{-t/RC} = i_0\, e^{-t/RC}$$

$$V = \frac{q}{C} = \frac{Qe^{-\frac{t}{RC}}}{C} = V_0 e^{-t/RC}$$



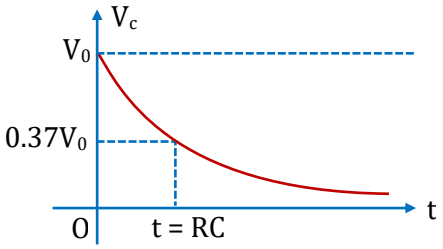


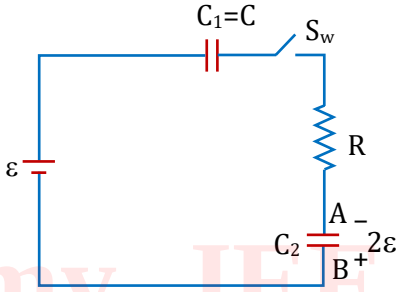
Illustration 46:

At t = 0, switch is closed, if initially C_1 is uncharged and C_2 is charged to a potential difference 2ε then find out following (Given $C_1 = C_2 = C$).

(a) Charge on C_1 and C_2 as a function of time.

(b) Find out current in the circuit as a function of time.

(c) Also plot the graphs for the relations derived in part (a).



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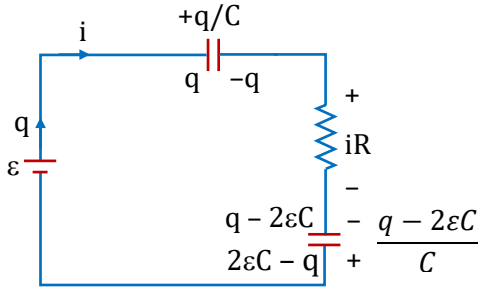
Solution:

Let q charge flow in time 't' from the battery as shown.

The charge on various plates of the capacitor is as shown in the figure.

Now applying KVL

$$\varepsilon - \frac{q}{C} - iR - \frac{q-2\varepsilon C}{C} = 0$$



$$\varepsilon - \frac{q}{C} - \frac{q}{C} + 2\varepsilon - iR = 0$$

$$3\varepsilon = \frac{2q}{C} + iR \Rightarrow 3\varepsilon - iR = \frac{2q}{C}$$

$$3\varepsilon C - iRC = 2q \quad \Rightarrow \quad \frac{dq}{dt}RC = 3\varepsilon C - 2q$$

$$\int_0^q \frac{dq}{3\varepsilon C - 2q} = \int_0^t \frac{dt}{RC} \quad \Rightarrow \quad -\frac{1}{2}\ell n\left(\frac{3C\varepsilon - 2q}{3C\varepsilon}\right) = \frac{t}{RC}$$

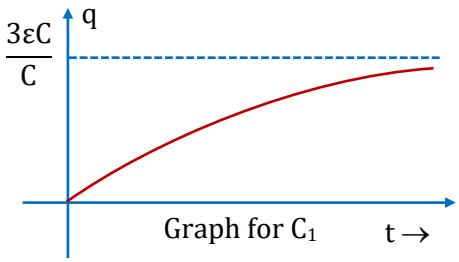
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$ln\left(\frac{3\varepsilon C-2q}{3\varepsilon C}\right) = -\frac{2t}{RC} \quad \Rightarrow \quad 3\varepsilon C - 2q = 3\varepsilon C e^{-2t/RC}$

$3\varepsilon C\,(1 - e^{-2t/RC}) = 2q \quad \Rightarrow \quad q = \frac{3}{2}\varepsilon C\,(1 - e^{-2t/RC})$

(charge on C, as function of time) **Ans.**

$i = \frac{dq}{dt} = \frac{3\varepsilon}{R}e^{-2t/RC}$ **Ans.**

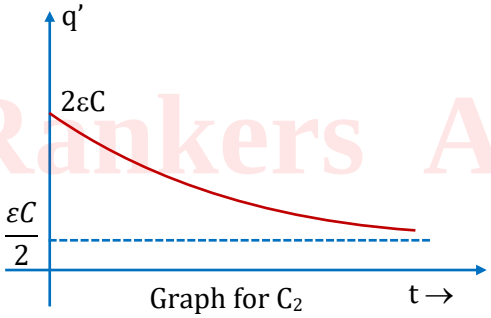


Charge on C_2 as function of time :

$q' = 2\varepsilon C - q$

$= 2\varepsilon C - \frac{3}{2}\varepsilon C + \frac{3}{2}\varepsilon C e^{-2t/RC} = \frac{\varepsilon C}{2} + \frac{3}{2}\varepsilon C e^{-2t/RC}$

$= \frac{\varepsilon C}{2}\left[1 + 3e^{-2t/RC}\right]$



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Illustration 47:

A 5.0 μF capacitor having a charge of 20 μC is discharged through a wire of resistance 5.0 Ω. Find the heat dissipated in the wire between 25 to 50 μs after the connections are made. (Given : $e^{-2} = 0.135$)

Solution:

$i_0 = \frac{20}{5} \times \frac{1}{5} = \frac{4}{5}$ amp.

$i = i_0 e^{-t/RC}$

$H = \int i^2 R dt = \int_{25\mu s}^{50\mu s} \frac{16}{5} e^{-2t/25\times10^{-6}} dt$

$= \frac{16}{5}\left(-\frac{25}{2}\times10^{-6}\right)\left[e^{-2t/25\times10^{-6}}\right]_{25\mu s}^{50\mu s}$

$= 40 \times 10^{-6}\left[\frac{e^2 - 1}{e^4}\right]$

$= 4.7\ \mu J$

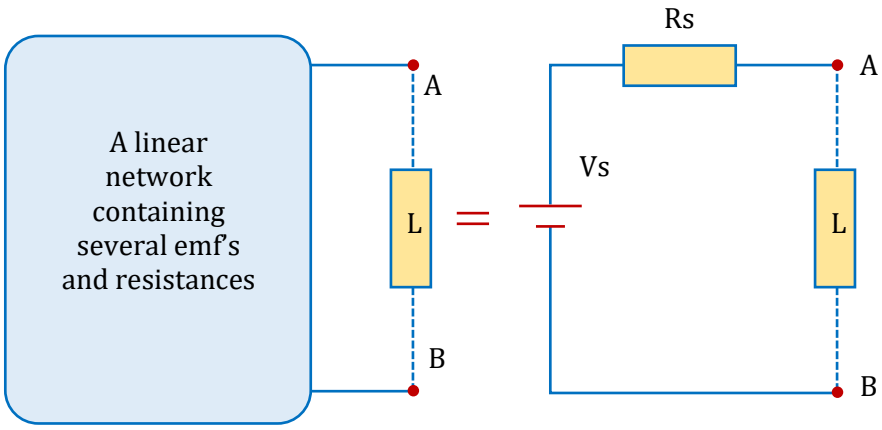
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Calculation of time constant of complex circuits

Thevenins Theorem:

To calculate time constant and write charge as function of time directly in those circuits where only one capacitor is present.

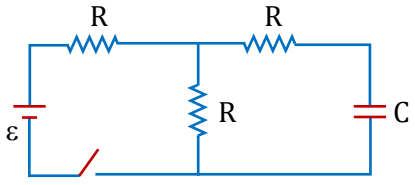
It states that any linear circuit containing any amount of emfs source and resistance can be replaced by a single voltage source called as Thevenins voltage (V_{Th}) and single series resistance connected to load which is termed as Thevenins resistance (R_{Th}).



In RC circuits we can take help of this method to simplify the circuit and calculate the time constant and charge on capacitor as function of time without actually analysing the circuit paths. In this method we have to simply open the circuit terminal where a capacitor is connected and replace the voltage sources by their internal resistances. Thus, the equivalent resistance between the terminals from where capacitor has been removed is terminals or effective resistance. The time constant hence can be written as $\tau = R_{eff.}\, C$.

In such cases if capacitor is initially uncharged we can directly use the result of charging RC circuit and write charge on capacitor at time t as $q = Q_{st}\left(1 - e^{-t/\tau}\right)$ where Q_{st} = steady state charge on capacitor.

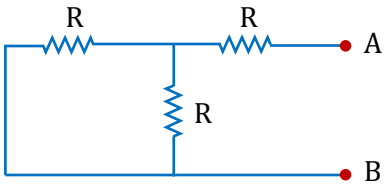
Let's try to understand the application of this method by illustrating following example



In this circuit we will try to find time constant and charge as a function of time taking help of Thevenins theorem. Calculation of $R_{\text{effective}}$.

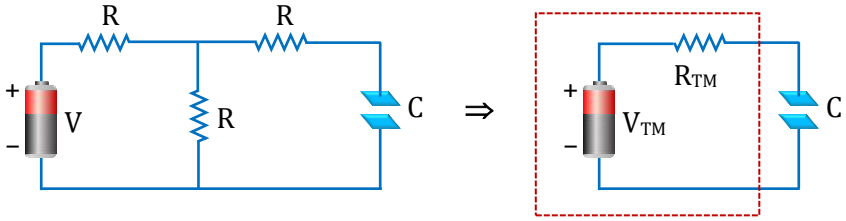
Therefore, Effective resistance will be equivalent of the combination between terminals A and B i.e. $3R/2$.

The given circuit will have time constant as $\tau = 3RC/2$.



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The given circuit can be redrawn as



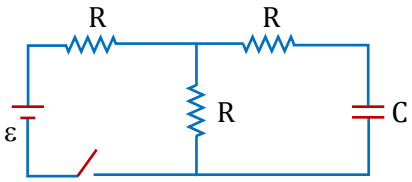
and we can write $q = Q_{st}\left(1 - e^{-t/\tau}\right)$; replacing time constant and steady state charge we will get the result of charge as function of time as $\Rightarrow q = \frac{\varepsilon C}{2}\left(1 - e^{-2t/3RC}\right)$.

Charging of complex RC circuits

There can be many complex RC circuit which may contain more than one resistor and capacitors. Charge as a function of time can be calculated by using KVL and KCL. This has been illustrated using KVL in simple following case.

Initially the capacitor is uncharged and here we will try to find the charge on capacitor as a function of time.

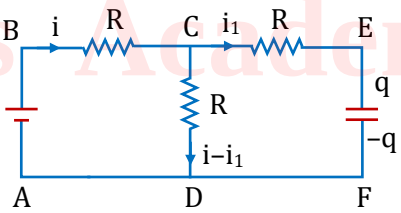
Let's assume that current through battery is i and it gets distributed in i_1 and $i - i_1$ as shown in figure. Also charge on capacitor at instant t is assumed to be q.



Applying KVL in loop ABCDA

$$\varepsilon - iR - (i - i_1)R = 0$$

$$\varepsilon - 2iR + i_1R = 0 \qquad \text{...(i)}$$



Applying KVL in loop ABCEFDA

$$\varepsilon - iR - i_1R - \frac{q}{C} = 0$$

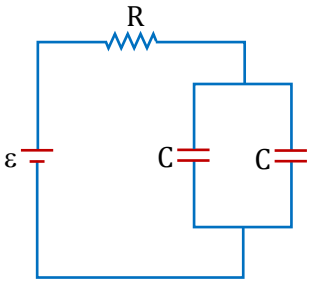
By equation (i)

$$\frac{2\varepsilon - \varepsilon - i_1R - 2i_1R}{2} = \frac{q}{C} \Rightarrow \varepsilon C - 3i_1RC = 2q$$

$$\varepsilon C - 2q = 3\frac{dq}{dt}.RC \Rightarrow \int_0^q \frac{dq}{\varepsilon C - 2q} = \int_0^t \frac{dt}{3RC}$$

$$-\frac{1}{2}\ln\frac{\varepsilon C - 2q}{\varepsilon C} = \frac{t}{3RC} \Rightarrow q = \frac{\varepsilon C}{2}\left(1 - e^{-2t/3RC}\right)$$

In this illustration we have more than one capacitor without using the formula of equivalent we will try to find out charge on capacitor and current in all the branches as a function of time.



Here we have assumed current through battery at any instant t as i which distributes as i_1 and i_2 in respectively capacitor branches as shown in diagram. Also, let's assume that charge at time t on capacitors C_1 and C_2 are q_1 and q_2 respectively.

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Applying KVL in ABDEA

$$\varepsilon - iR = \frac{q}{2C}$$

$$i = \frac{\varepsilon}{R} - \frac{q}{2CR} = \frac{2C\varepsilon - q}{2CR}$$

$$\frac{dq}{2\varepsilon C - q} = \frac{dt}{2CR}$$

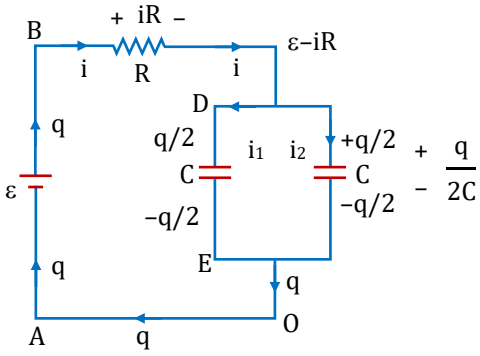
$$\int_0^q \frac{dq}{(2\varepsilon C - q)} = \frac{t}{2CR}$$

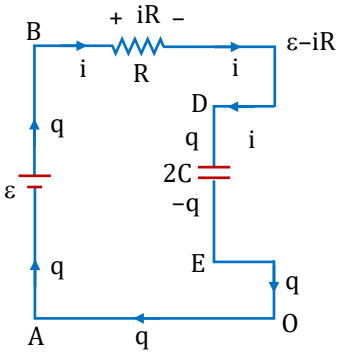
$$\frac{2\varepsilon C - q}{2\varepsilon C} = e^{-t/2RC}$$

$$q = 2\varepsilon C\,(1 - e^{-t/2RC})$$

$$q_1 = \frac{q}{2} = \varepsilon C\,(1 - e^{-t/2RC}) \quad \Rightarrow i_1 = \frac{\varepsilon}{2R} e^{-t/2RC}$$

$$q_2 = \frac{q}{2} = \varepsilon C\,(1 - e^{-t/2RC}) \quad \Rightarrow i_2 = \frac{\varepsilon}{2R} e^{-t/2RC}$$





Alternate solution:

By equivalent

Time constant of circuit = $2C \times R = 2RC$

Maximum charge on capacitor = $2C \times \varepsilon = 2C\varepsilon$

Hence equations of charge and current are as given below

$$q = 2\varepsilon C\,(1 - e^{-t/2RC})$$

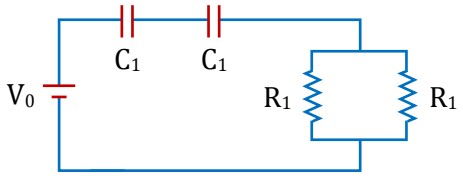
$$q_1 = \frac{q}{2} = \varepsilon C\,(1 - e^{-t/2RC}) \quad \Rightarrow i_1 = \frac{\varepsilon}{2R} e^{-t/2RC}$$

$$q_2 = \frac{q}{2} = \varepsilon C\,(1 - e^{-t/2RC}) \quad \Rightarrow i_2 = \frac{\varepsilon}{2R} e^{-t/2RC}$$

Illustration 48:

Find the time constant given circuit if

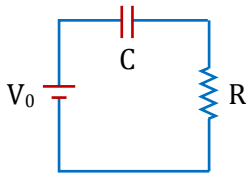
$R_1 = 4\Omega$, $R_2 = 12\Omega$, $C_1 = 3\mu F$ and $C_2 = 6\mu F$.



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Solution:

Given circuit can be reduced to



$$C = \frac{C_1C_2}{C_1+C_2} = \frac{3\times6}{3+6} = 2\mu F, \quad R = \frac{R_1R_2}{R_1+R_2} = \frac{4\times12}{4+12} = 3\Omega$$

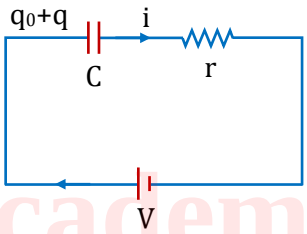
Time constant = RC = (3)(2) = 6 μs.

Illustration 49:

A capacitor of capacitance C is charged by charge q_0. At t = 0, it is connected to a battery of emf V and internal resistance r. Find the charge on the capacitor at time t (positive plate of capacitor connected with positive plate of battery).

Solution:

At time t using KVL



$$-\left(\frac{q_0+q}{C}\right) - ir + V = 0$$

$$\Rightarrow \quad -\left(\frac{q_0+q}{C}\right) - \frac{dq}{dt}r + V = 0$$

$$\Rightarrow \quad \int_0^q \frac{dq}{(CV-(q_0+q)} = \int_0^t \frac{dt}{rC}$$

By using integration,

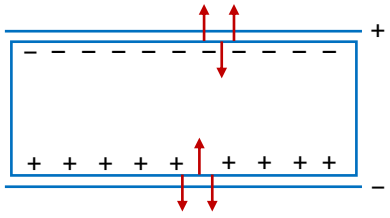
$$q = CV\,(1 - e^{-t/RC}) + q_0\,e^{-t/RC} - q_0$$

$$\Rightarrow \quad \text{So, charge on capacitor} = q_0 + q = CV\,(1 - e^{-t/RC}) + q_0e^{-t/RC}$$

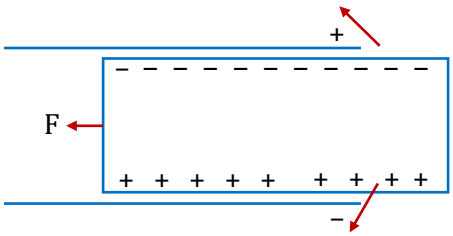
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Force on a dielectric due to charged capacitor

If dielectric is completely inside the capacitor then force is equal to zero.



If dielectric is not completely inside the capacitor.



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Case I : Voltage source remains connected

V = constant.

$U = \frac{1}{2}CV^2$

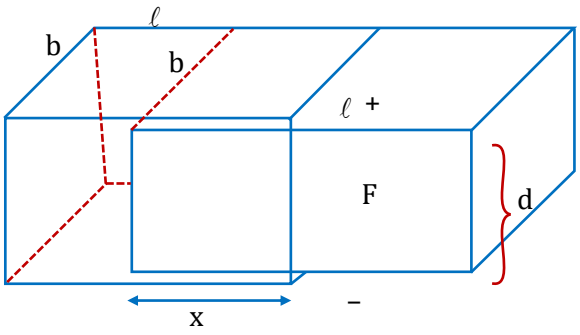
$|\vec{F}| = \left(\frac{dU}{dx}\right) = \frac{V^2}{2}\frac{dC}{dx}$

where $C = \frac{xb\epsilon_0 K}{d} + \frac{\epsilon_0(\ell - x)b}{d}$

$\Rightarrow C = \frac{\epsilon_0 b}{d}[Kx + \ell - x]$

$\frac{dC}{dx} = \frac{\epsilon_0 b}{d}(K-1)$

$\therefore \quad F = \frac{\epsilon_0 b(K-1)V^2}{2d}$ = constant (does not depend on x)



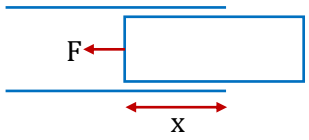
Case II : When charge on capacitor is constant

$C = \frac{xb\epsilon_0 K}{d} + \frac{\epsilon_0(\ell - x)b}{d}, U = \frac{Q^2}{2C}$

$F = \left(\frac{dU}{dx}\right) = \frac{Q^2}{2C^2} \cdot \frac{dC}{dx}$ [where, $\frac{dC}{dx} = \frac{\epsilon_0 b}{d}(K-1)$]

$F = \frac{Q^2}{2C^2} \times \frac{\epsilon_0 b}{d}(K-1) = \frac{\varepsilon_0 b(k-1)Q^2}{2dC^2}$

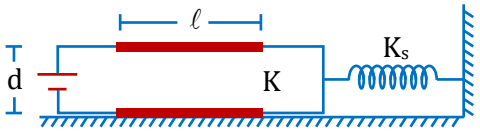
(here force 'F' depends on x)



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Illustration 50:

Consider the situation shown in figure. The width of each plate is b. The capacitor plates are rigidly clamped in the laboratory and connected to a battery of emf V. All surface are frictionless. Calculate the extension in the spring in equilibrium (spring is nonconducting).



Solution:

Potential energy $= \frac{1}{2}(C_1 + C_2)V^2$

$= \frac{1}{2}\left\{\frac{\epsilon_o xb}{d} + \frac{\epsilon_o K(\ell - x)b}{d}\right\} V^2$

$F = -\frac{\partial U}{\partial x} = -\frac{V^2}{2}\frac{dC}{dx}$

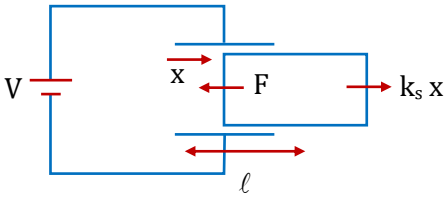
$F = \frac{-1}{2}\frac{dC}{dx}V^2$

where, $\frac{dC}{dx} = \frac{\epsilon_o b}{d}\{1 - K\}$

$F = \frac{1}{2}\frac{V^2 \epsilon_o b(K-1)}{d}$

At equilibrium, $F = K_s x$

$x = \frac{F}{k_S} = \frac{\epsilon_0 bV^2(K-1)}{2dK_S}$



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